

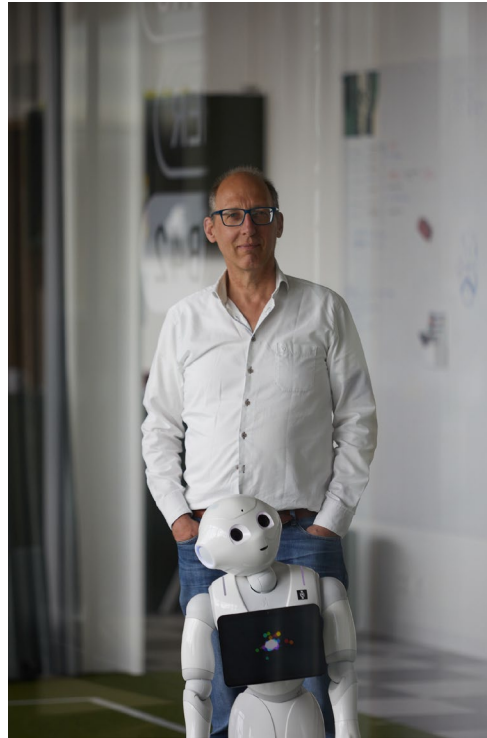
Computer Vision 1

Master of AI – 2024/2025
University of Amsterdam

Arnoud Visser & Martin Oswald

Based on a presentation of Shaodi You,
augmented with material of Linda Shapiro & many others

A little bit of my background



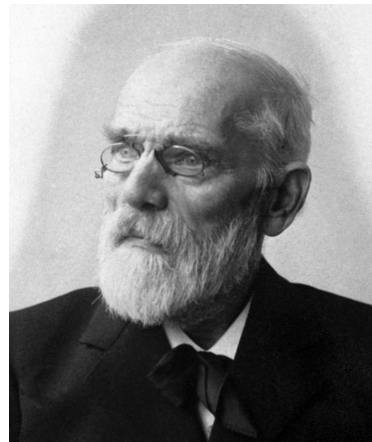
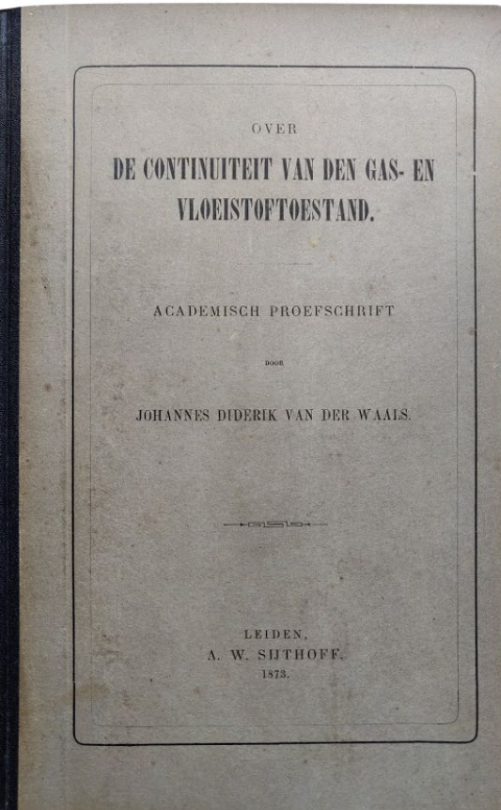
Arnoud Visser

PhD in Computer Science,

Master in Physics, Minor in BioChemistry



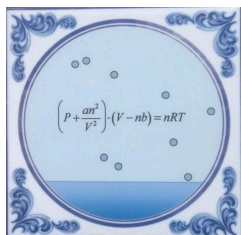
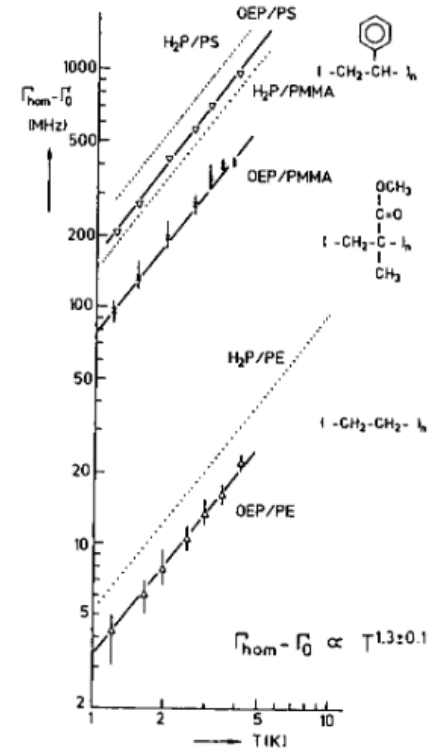
Education



*Prof. Dr. Johannes Diderik
van der Waals
(1837-1923)*



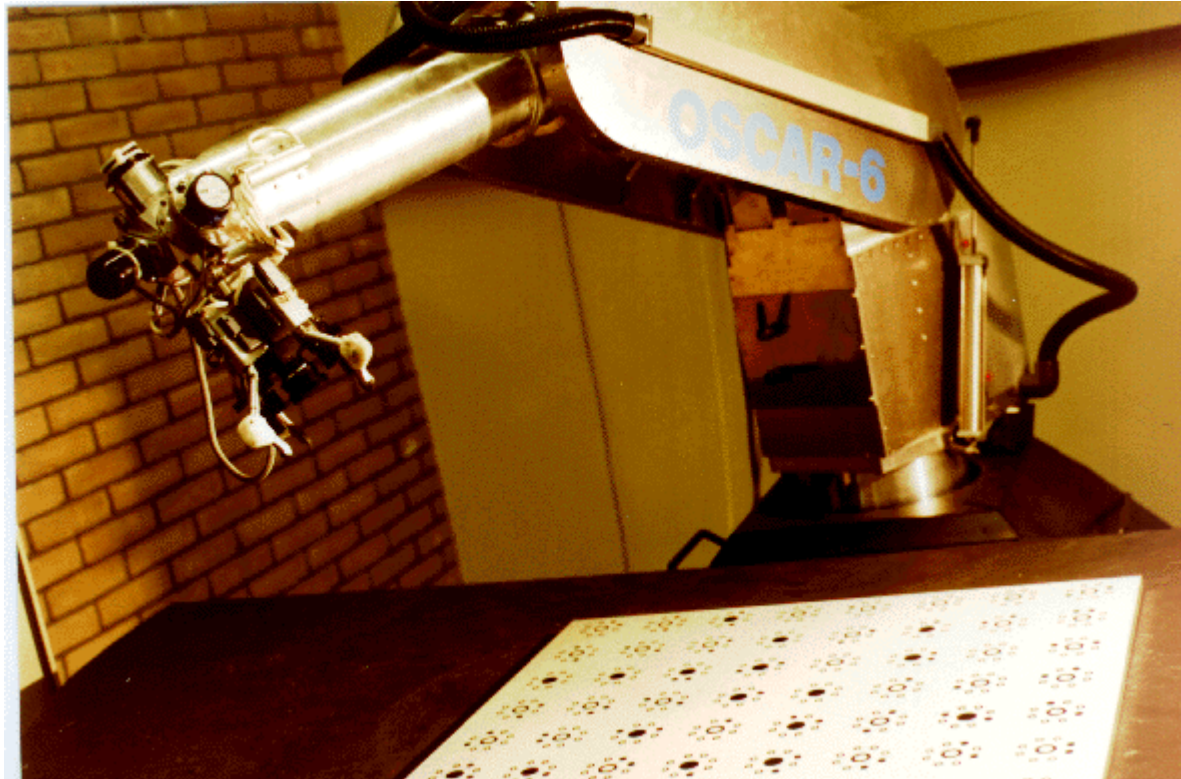
*Prof. Dr. Joan Henri
van der Waals
(1920-2022)*



PhD student in Experimental Physics



Robot arms



Esprit-CAR

Calibration Applied to Quality Control and Maintenance in Robot Production
1991-1994

Flying robots



Space robots



Soccer playing robots

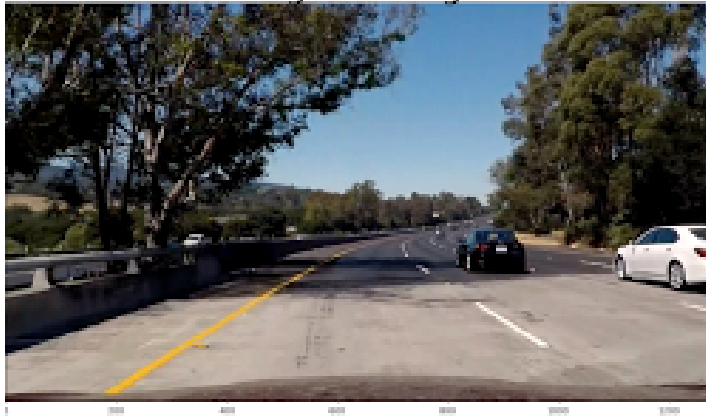


Humanoid robots

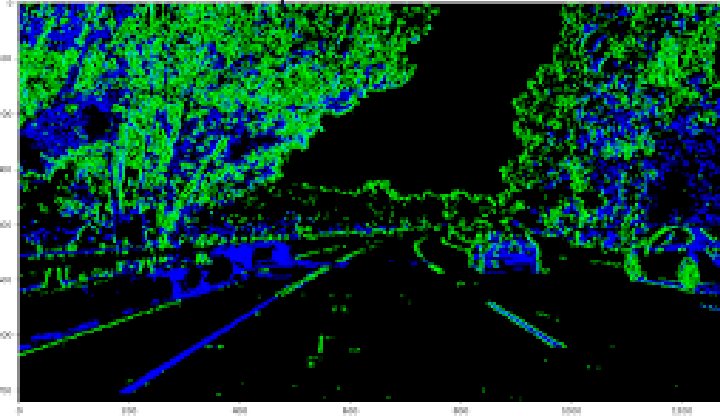


Perception for Mobile Robots

Original Image



Pipeline Result



robot calibration

object recognition

virtual sensors

localization and mapping

exploration

So, what is Computer Vision ?

Let's poll

Question 1

I have learnt computer vision in my bachelor courses.

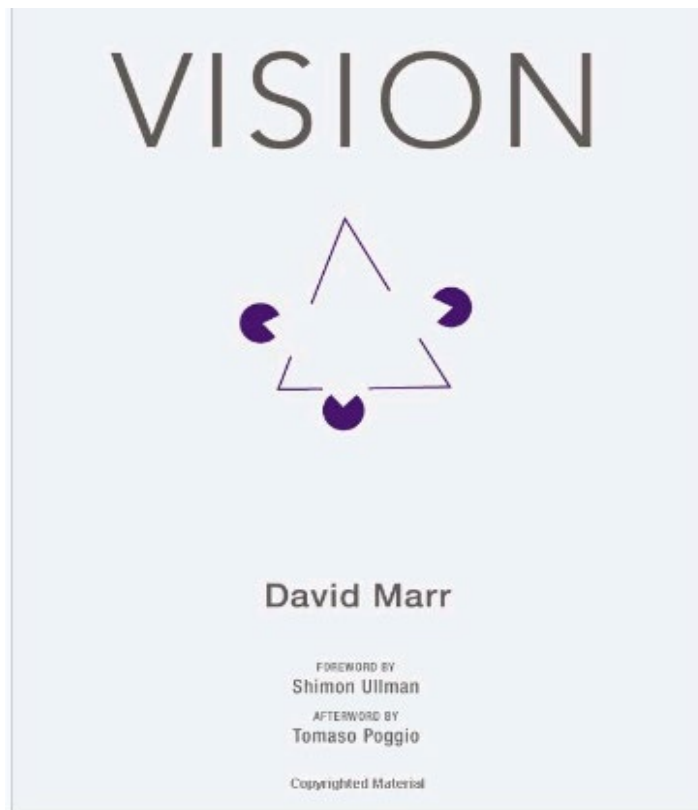
Question 2

I am a UvA student and finished the bachelor course
Introduction to Computer Vision

What is Vision?

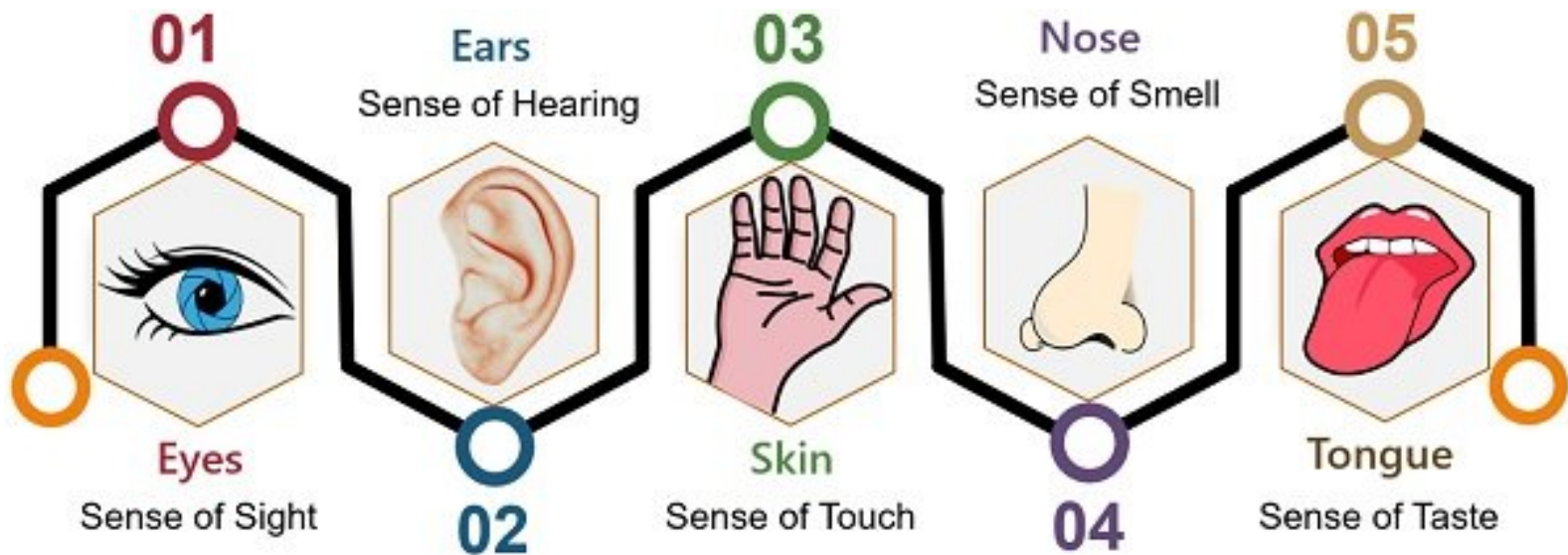
- Is [the art] *“to know what is where, by looking.”*

David Marr (1945 --1980).



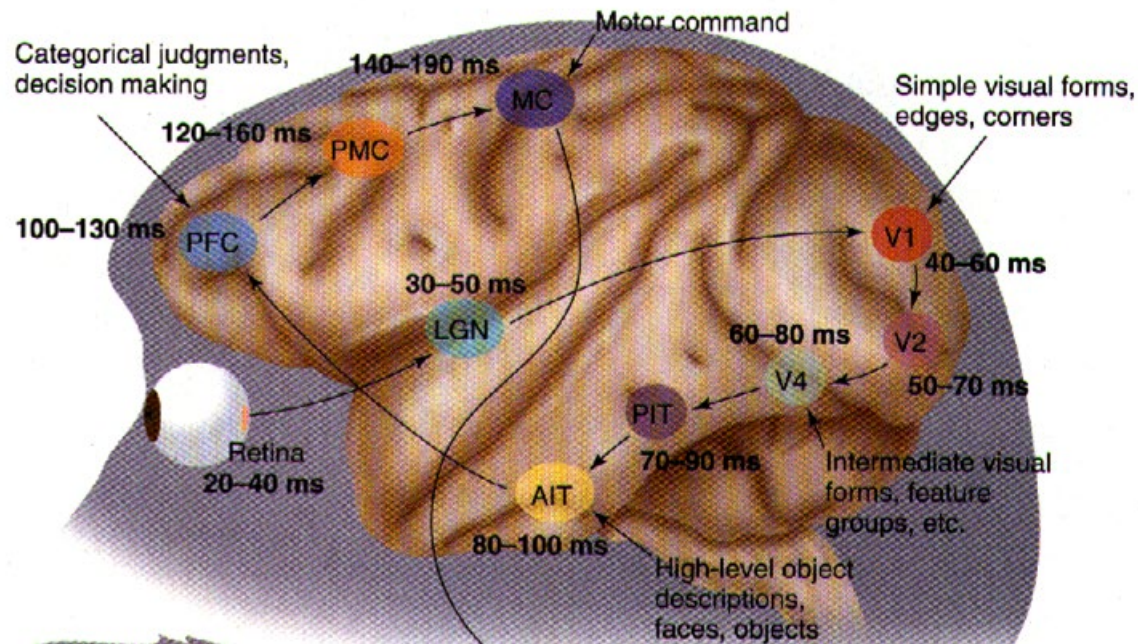
Why study vision in AI

Vision is the major perception of the five human senses



Human visual system

- Vision is the most powerful of our senses.

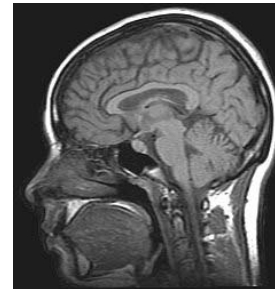


[Thorpe et. al.]

- Around 1/3 of our brain is devoted to processing the signals from our eyes.
- The visual cortex alone has around $O(10^{11})$ neurons.

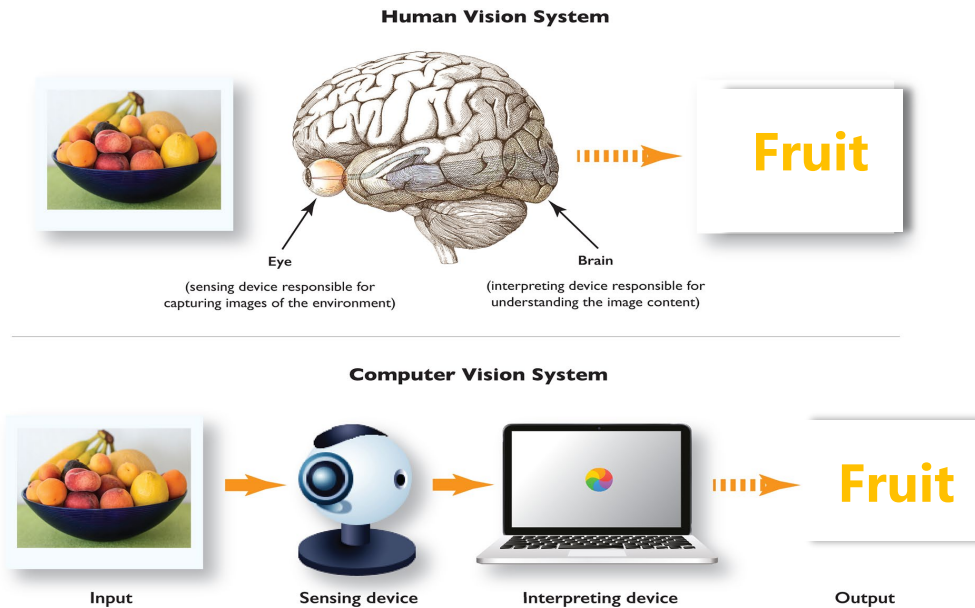
What is Computer Vision?

- Computer Vision is concerned with the theory of building artificial vision systems that obtain useful information from images.
- The image data can take many forms, such as a video sequence, depth images, views from multiple cameras, or multi-dimensional data from a medical scanner.



The computer vision problem

- Make a computer see and understand images



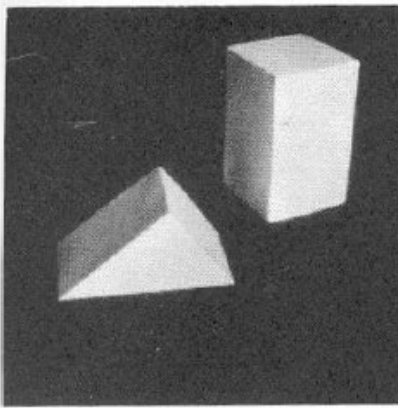
- We know it is physically possible –
we do it every day and effortlessly!

Brief history of computer vision research

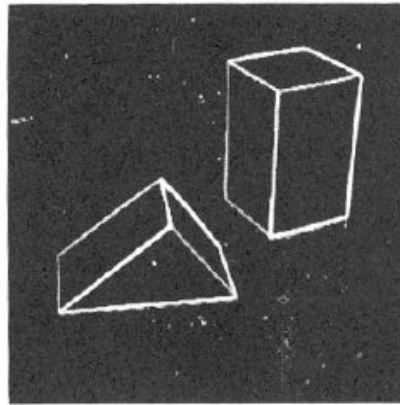


Electrologica X8 computer (1964), Boerhaave museum, Leiden

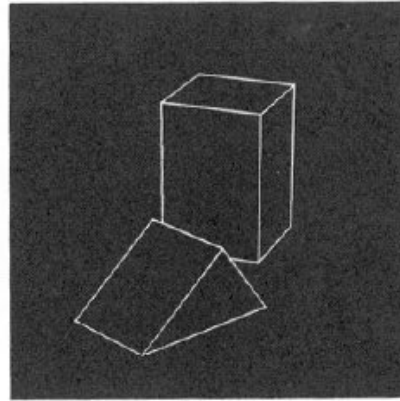
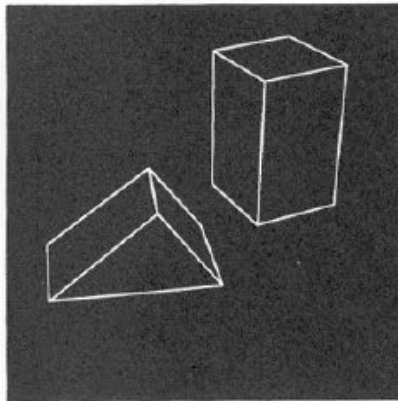
Origins of Computer Vision



(a) Original picture.



(b) Differentiated picture.



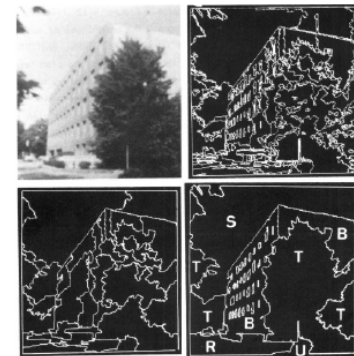
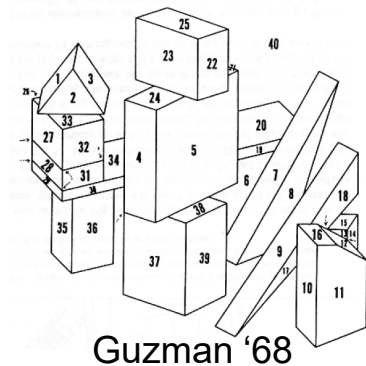
L. G. Roberts, *Machine Perception of Three Dimensional Solids*, Ph.D. thesis, MIT Department of Electrical Engineering, 1963.

An (overly) optimistic start

- In 1966, Marvin Minsky at MIT asked his undergraduate student Gerald Jay Sussman to “spend 3 months in this summer linking a camera to a computer and getting the computer to describe what it saw”.
- Now, over fifty years, we know the problem is significantly more difficult than a 3-month student project.

Brief history of computer vision

- 1963: Robert's thesis
- 1966: Minsky assigns computer vision as an undergrad summer project
- 1960's: interpretation of synthetic worlds
- 1970's: some progress on interpreting selected images
- 1980's: ANNs come and go; shift toward geometry and increased mathematical rigor, Marr's book published (after he died).
- 1990's: face recognition; statistical analysis in vogue
- 2000's: broader recognition; large annotated datasets available; video processing starts; vision & graphics; vision for HCI; internet vision, etc.
- 2012: Kinect, big-data, Google Car, ...
- 2013~: Deep learning, Image-Net and ...
- 2020~: Transformer, Vision and Language, diffusion model ...

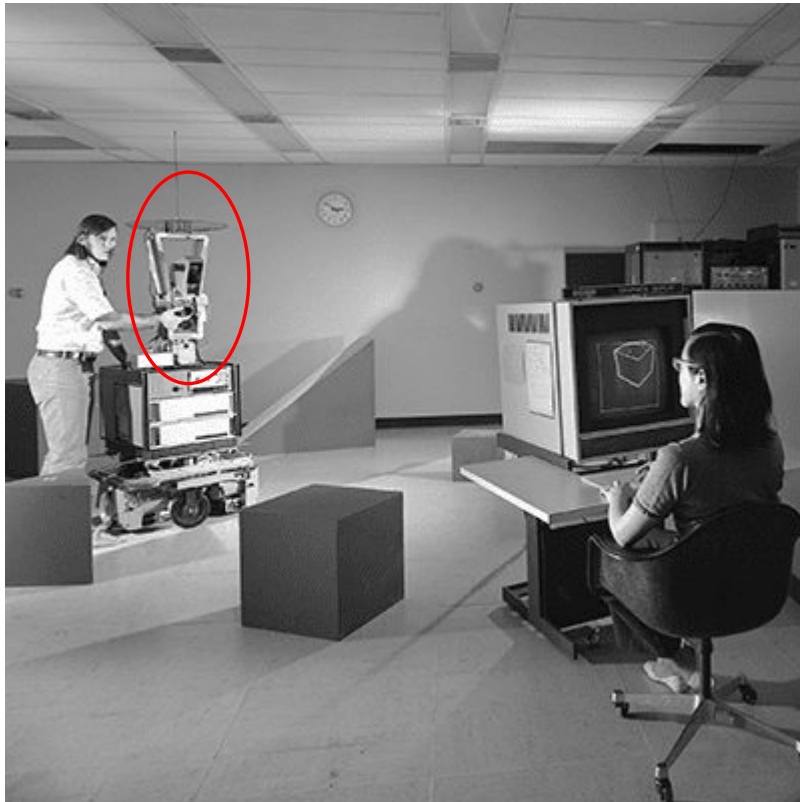


Ohta Kanade '78



Turk and Pentland '91²¹

Starts with a good sensing device



Early AI robot: Shakey 1969

- Visual navigation in special world
- Vision and contact sensors



1917
The birth of Nikon



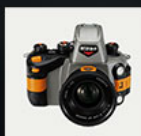
1933
Aero-NIKKOR
First NIKKOR lenses



1957
Nikon SP
World's first camera with built-in 28mm-135mm universal finder



1979
Nikon EM
First ultra-compact SLR



1992
NIKONOS RS
World's first underwater SLR



2008
Nikon D90
World's first DSLR with Video Capture



2016
Nikon D5



1952
Founding of NIKKOR Club



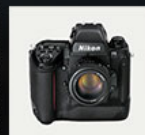
1963
NIKONOS
First all-weather camera



1988
Nikon F4
World's first camera with vertical sensor for metering
Nippon Kogaku K.K. becomes Nikon Corporation



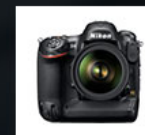
1994
Nikon Zoom 700VR/700VR QD
World's first compact with Vibration Reduction



1996
Nikon F5
World's first 1005-pixel RGB sensor for metering



2003
New Nikon Brand Symbol



2012
Nikon D4

2016
100 million lenses produced



1929
Anytar
First photographic lens



1948
Nikon Model I
First compact Nikon RF camera



1959
Nikon F
First interchangeable lens SLR camera
Nikon F mount

1969
First Nikon Photo Contest



Nikon F2



Development of Extra-Low Dispersion Glass



1980
Nikon F3

1986



Nikon F-501
First SLR with autofocus operated from camera body
Nikkor 50mm f/1.8S
World's first Lens with built-in CPU

1983



Nikon L35AF2 (One-Touch)
First autofocus compact camera
Development of AF mechanism



1997
COOLPIX 100
First compact digital camera



1999
Nikon D1 DSLR
First DSLR for consumers



Nikon F6
Last of the Nikon F-Series

2004



CoolPix P1
World's first compact camera with WIFI

2005



Nikon 1 AW1
World's first waterproof digital camera

2013



2016
KeyMission 360, KeyMission 170, and KeyMission80
First action cameras



Opening of the Nikon Museum

Camera and Camera Module Overview

Front Single Cameras:

Front Modules:

module retained

Front

Front

Front

Front

Front NIR

Front NIR

Front NIR

Front NIR

module retained

TrueDepth Front Camera:



iPhone 6s Plus

iPhone 7 Plus

iPhone 8 Plus

iPhone X

iPhone Xs Max

iPhone 11 Pro Max

iPhone 12 Pro Max

Rear Single Camera:



Rear Dual Cameras:



Rear Triple Cameras:



Rear Modules:



Wide

Telephoto



Wide

Telephoto



Wide



Telephoto

module retained



Wide



Telephoto

module retained

Main



Telephoto

Ultra Wide



Telephoto

Ultra Wide

SAMPLE

Multiple camera's, multiple foci



Light L16 (2018)
a multi-camera device that
computationally creates
high-quality, high-resolution
photos from up to 16 small,
low-resolution sensors



Hyper resolution images

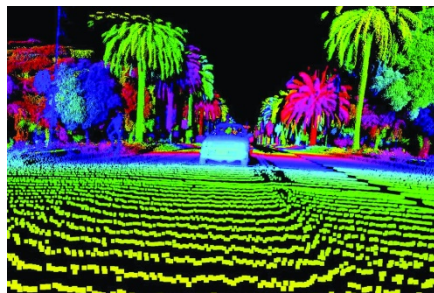


Operation Night Watch made the largest and most detailed photo ever taken of a work of art. It is 717 gigapixels, or 717,000,000,000 pixels, in size.

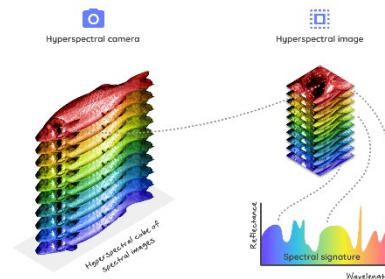
The distance between two pixels is 5 micrometers, which means that one pixel is smaller than a human red blood cell.

The team used a 100-megapixel Hasselblad H6D 400 MS-camera to make 8439 individual photos measuring 5.5cm x 4.1cm. Artificial intelligence was used to stitch these smaller photographs together to form the final large image, with a total file size of 5.6 terabytes.

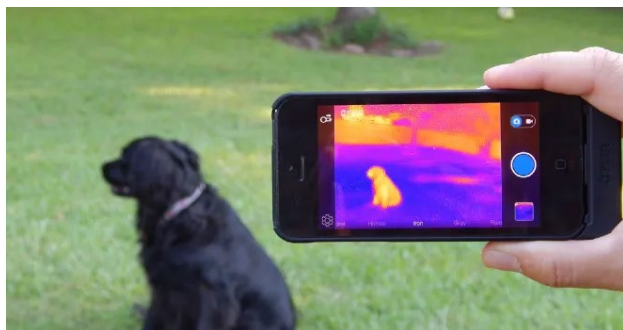
Many other vision sensors



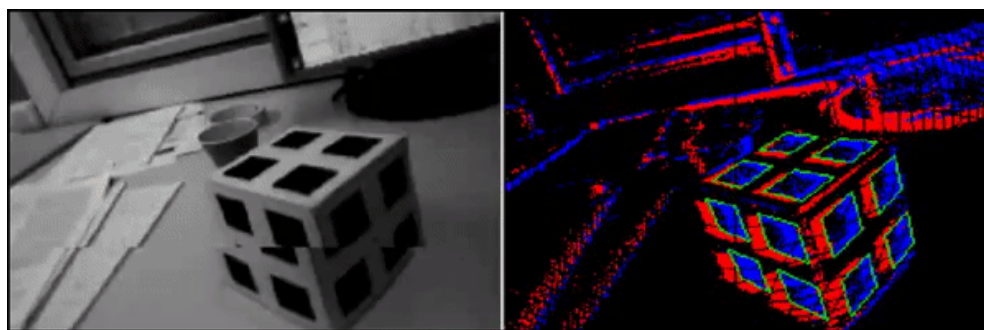
Lidar



Hyperspectral



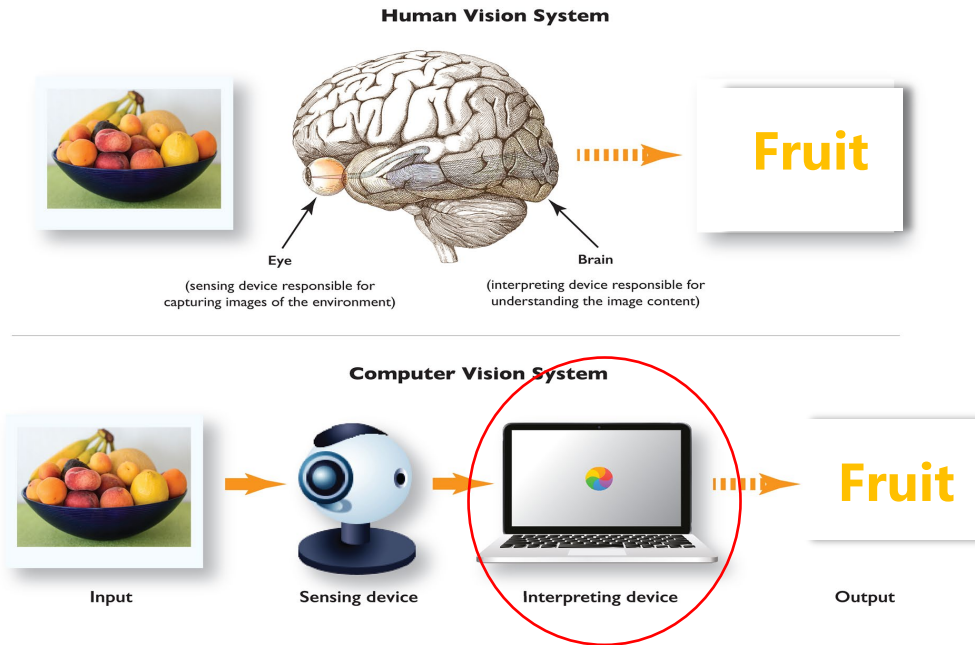
Infra Red



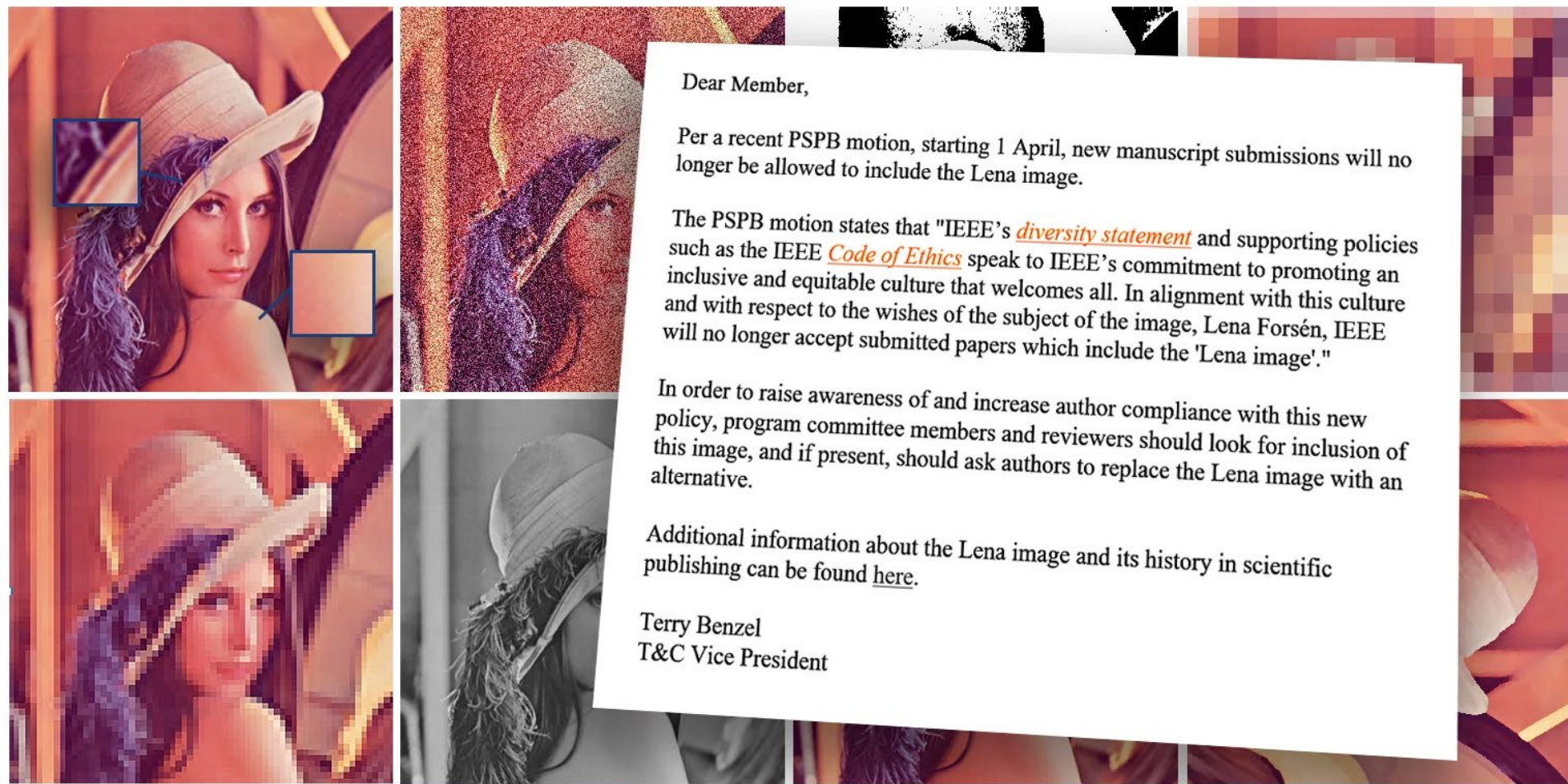
Event camera

The computer vision problem

- Make a computer see and understand images



- We know it is physically possible –
we do it every day and effortlessly!



Dear Member,

Per a recent PSPB motion, starting 1 April, new manuscript submissions will no longer be allowed to include the Lena image.

The PSPB motion states that "IEEE's *diversity statement* and supporting policies such as the IEEE *Code of Ethics* speak to IEEE's commitment to promoting an inclusive and equitable culture that welcomes all. In alignment with this culture and with respect to the wishes of the subject of the image, Lena Forsén, IEEE will no longer accept submitted papers which include the 'Lena image'."

In order to raise awareness of and increase author compliance with this new policy, program committee members and reviewers should look for inclusion of this image, and if present, should ask authors to replace the Lena image with an alternative.

Additional information about the Lena image and its history in scientific publishing can be found [here](#).

Terry Benzel
T&C Vice President

Lena Forsén Picture Used as Reference Photo

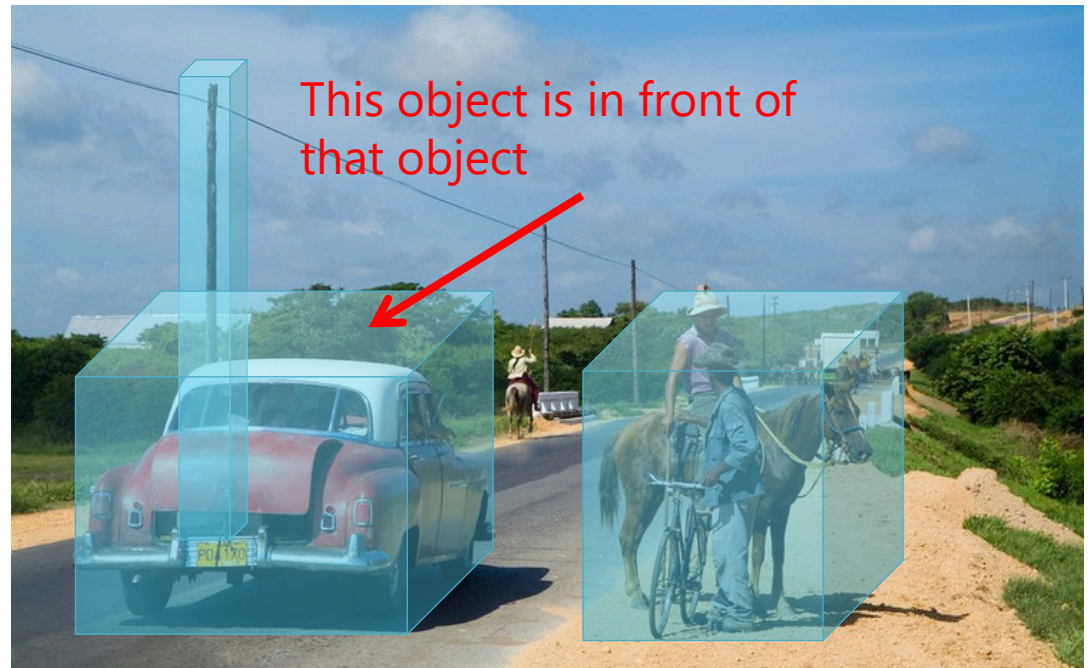
Computer Vision

- Low Level Vision
 - Measurements
 - Enhancements
 - Region segmentation
 - Features
- Mid Level Vision
 - Reconstruction
 - Depth
 - Motion Estimation
- High Level Vision
 - Category detection
 - Activity recognition
 - Deep understandings



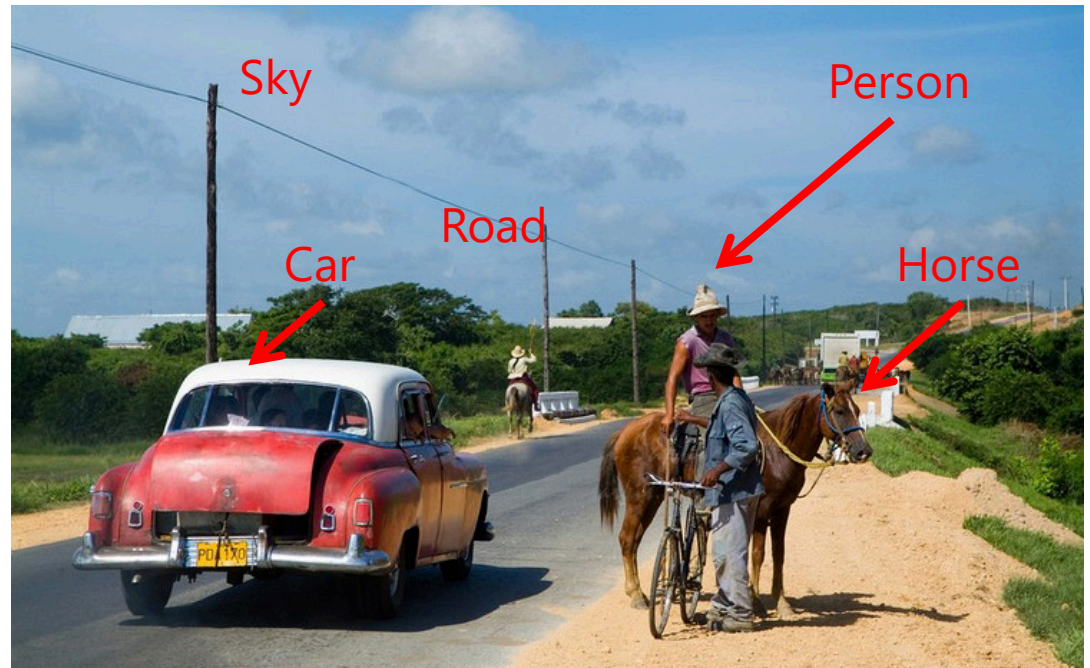
Computer Vision

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 - Category detection
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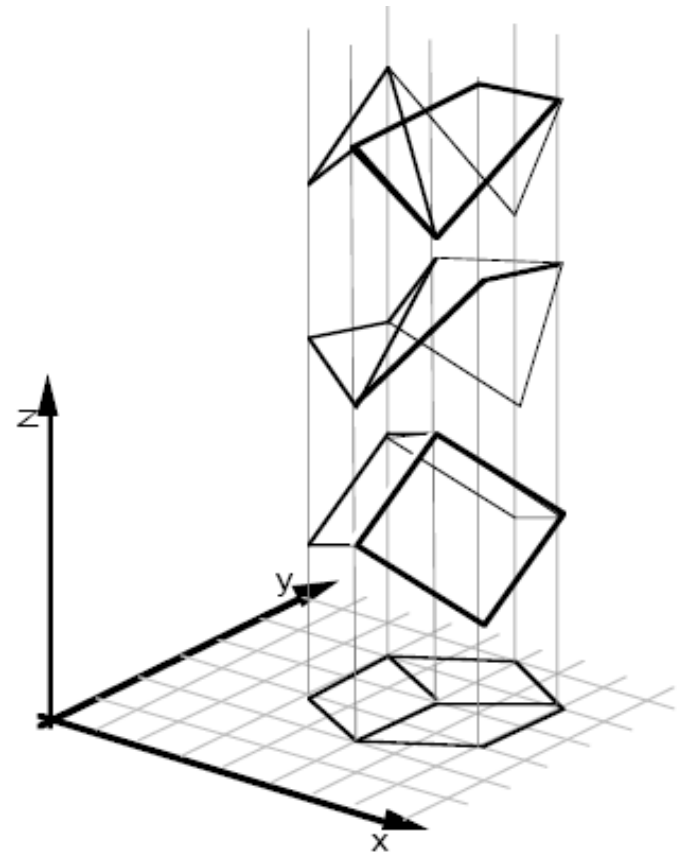
Computer Vision

- Low Level Vision
 - Measurements
 - Enhancements
 - Region segmentation
 - Features
- Mid Level Vision
 - Reconstruction
 - Depth
 - Motion Estimation
- High Level Vision
 - Category detection
 - Activity recognition
 - Deep understandings
 - Pose estimation



Why is vision so hard?

- Ill-posed problem



[Sinha and Adelson 1993]

Challenges 1: view point variation



Challenges 2: illumination



Challenges 3: occlusion



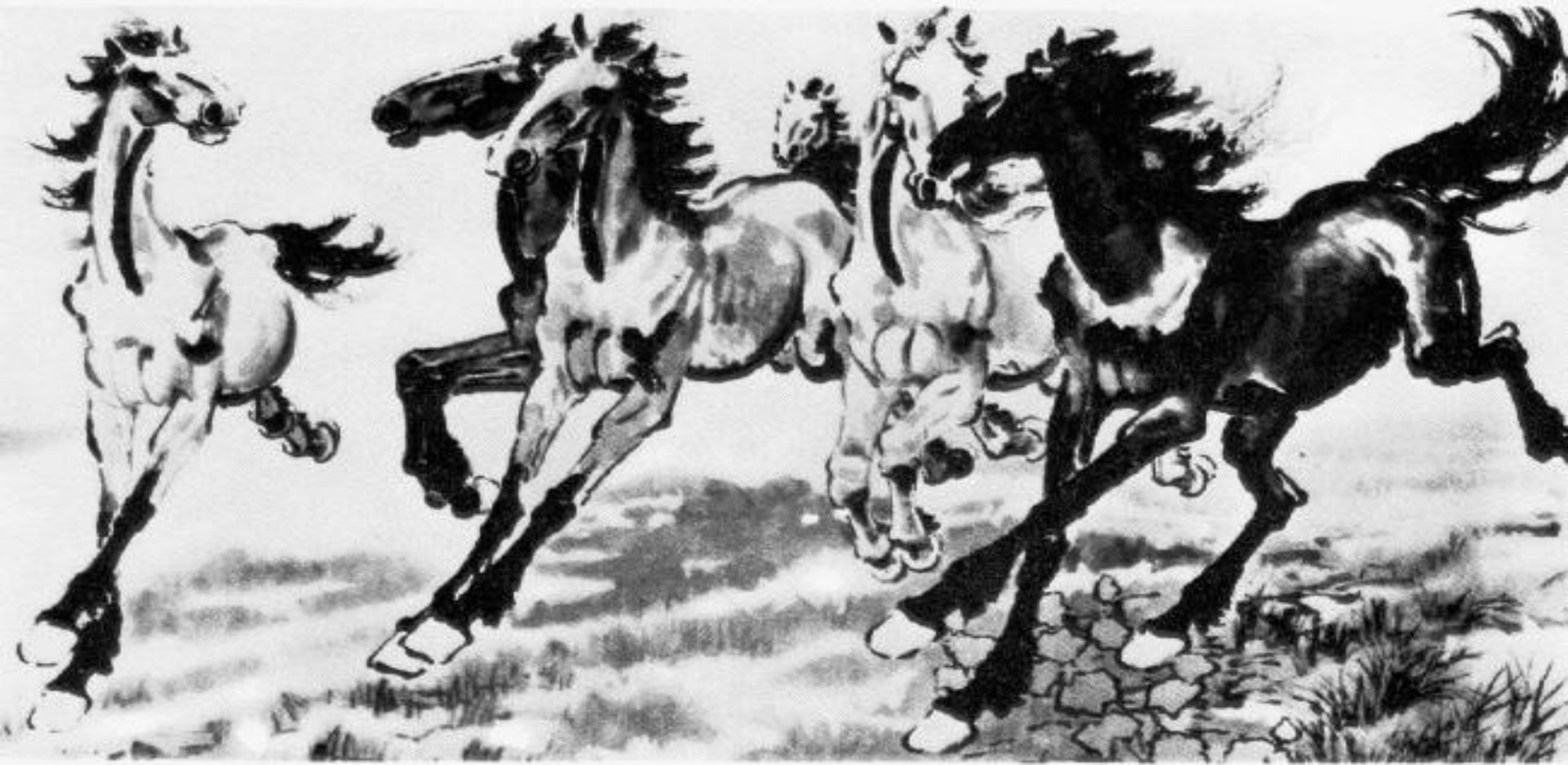
Magritte, 1957

slide by Fei Fei, Fergus & Torralba³⁷

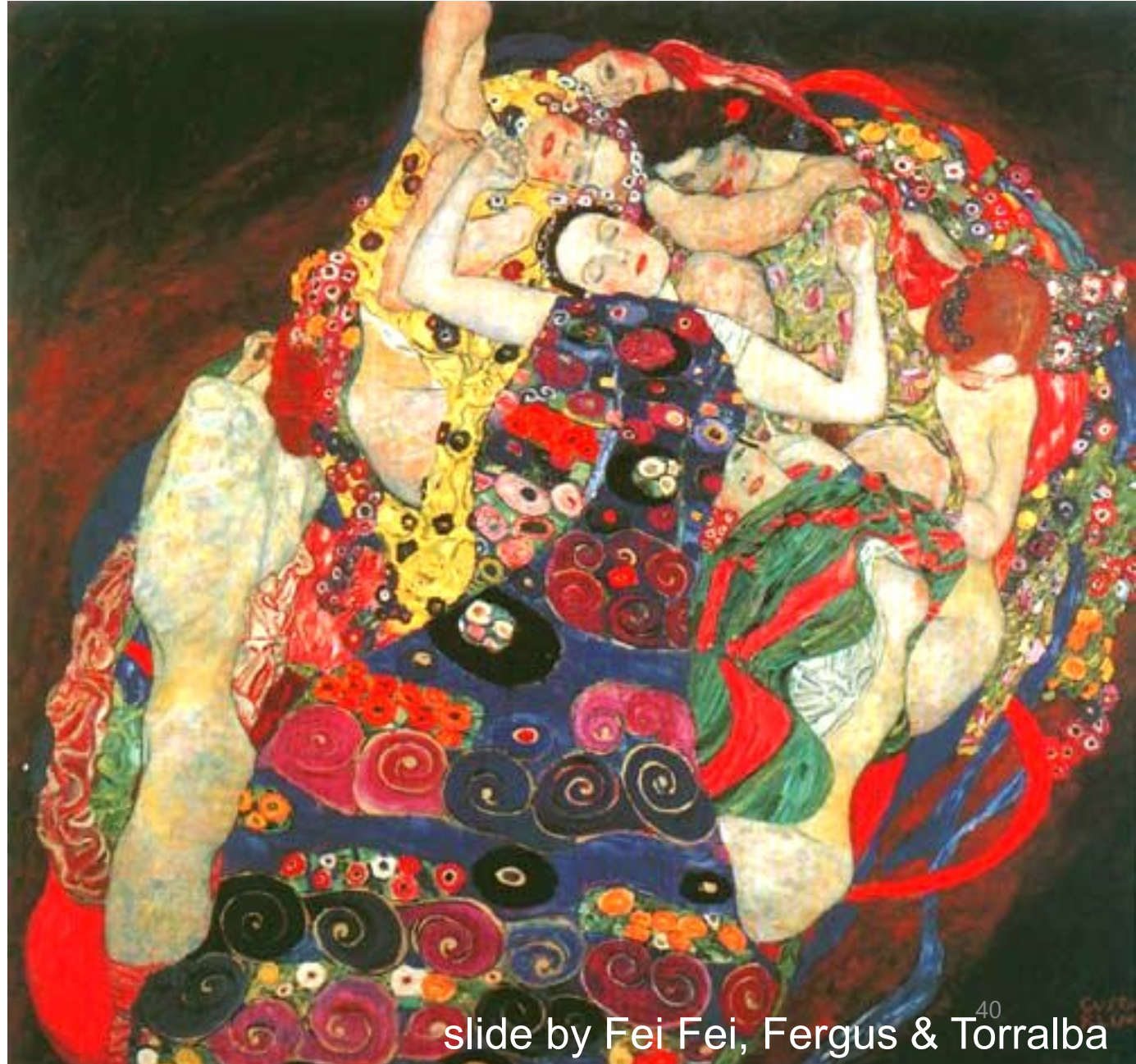
Challenges 4: scale



Challenges 5: deformation



Challenges 6: background clutter

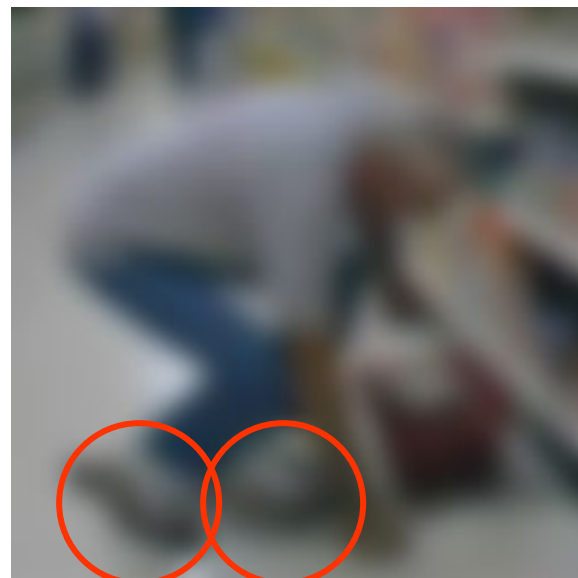
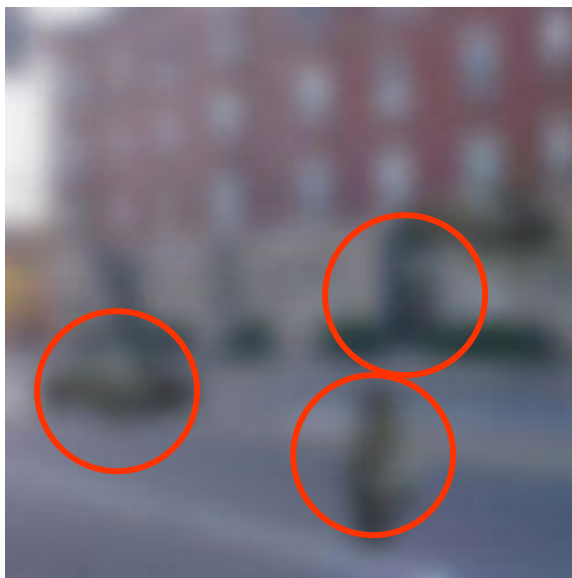
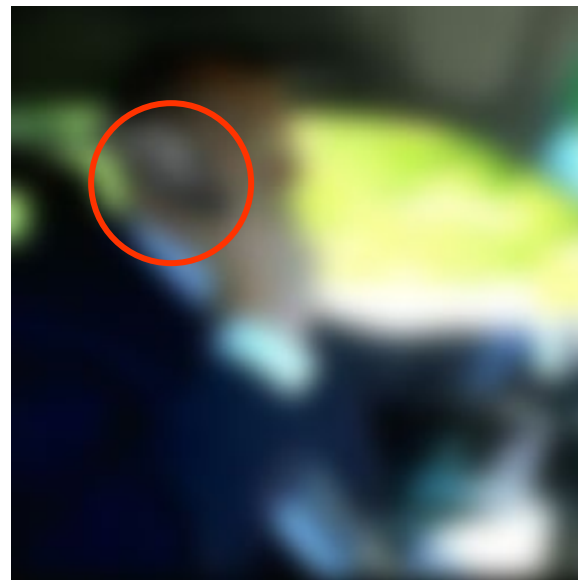
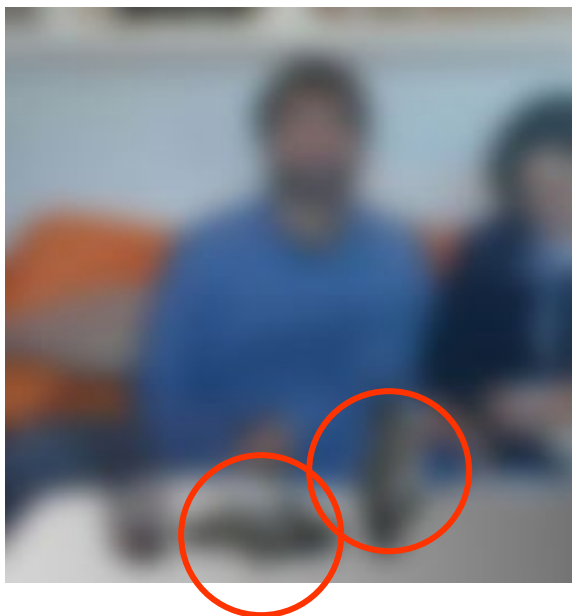
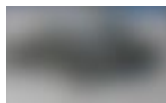


Klimt, 1913

Challenges 7: object intra-class variation



Challenges 8: local ambiguity



Challenges 9: the world behind the image



Artificial Intelligence

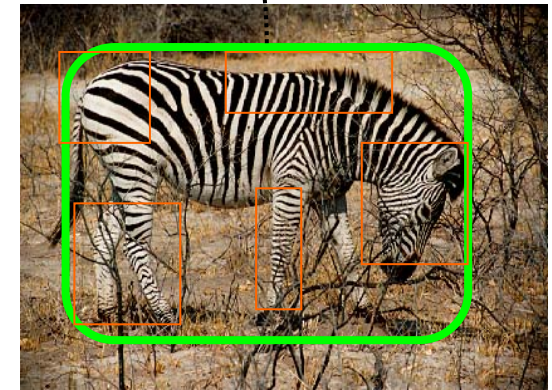
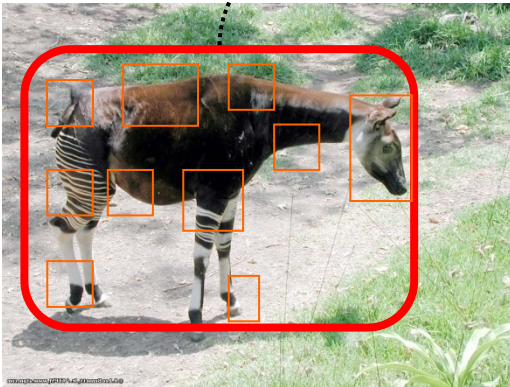
Machine learning

- Direct modeling of $\frac{p(\text{zebra} \mid \text{image})}{p(\text{no zebra} \mid \text{image})}$

Decision
boundary

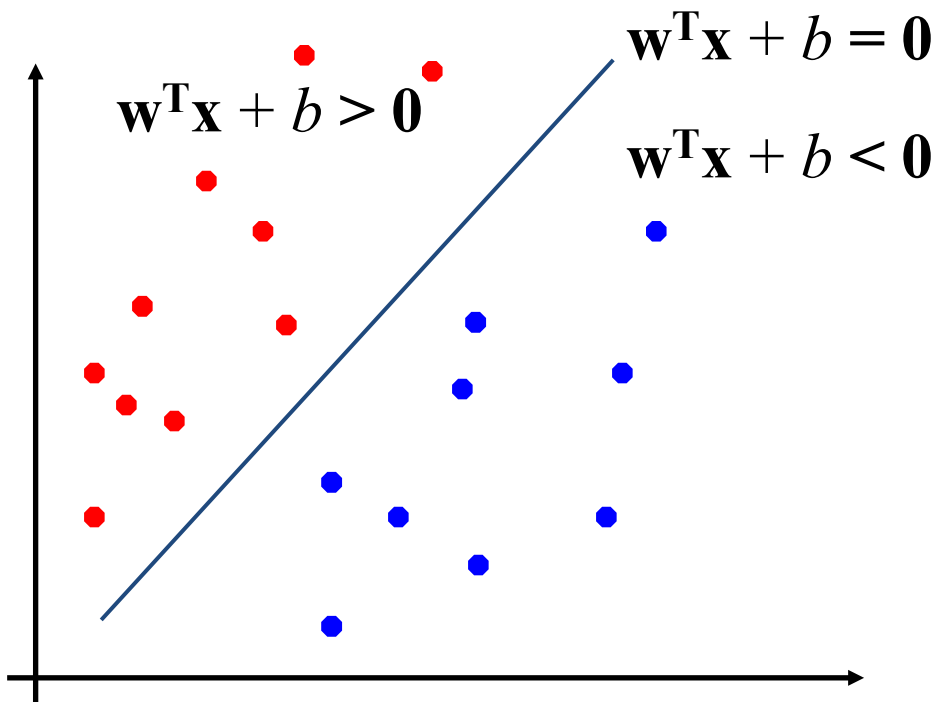
Zebra

Non-zebra



Linear Separators

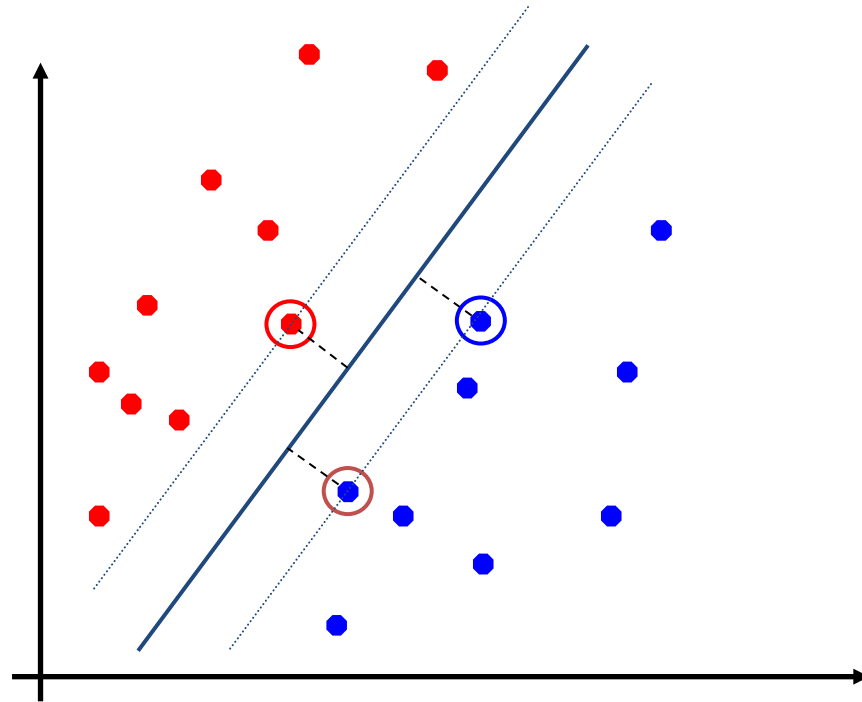
Binary classification can be viewed as the task of separating classes in feature space:



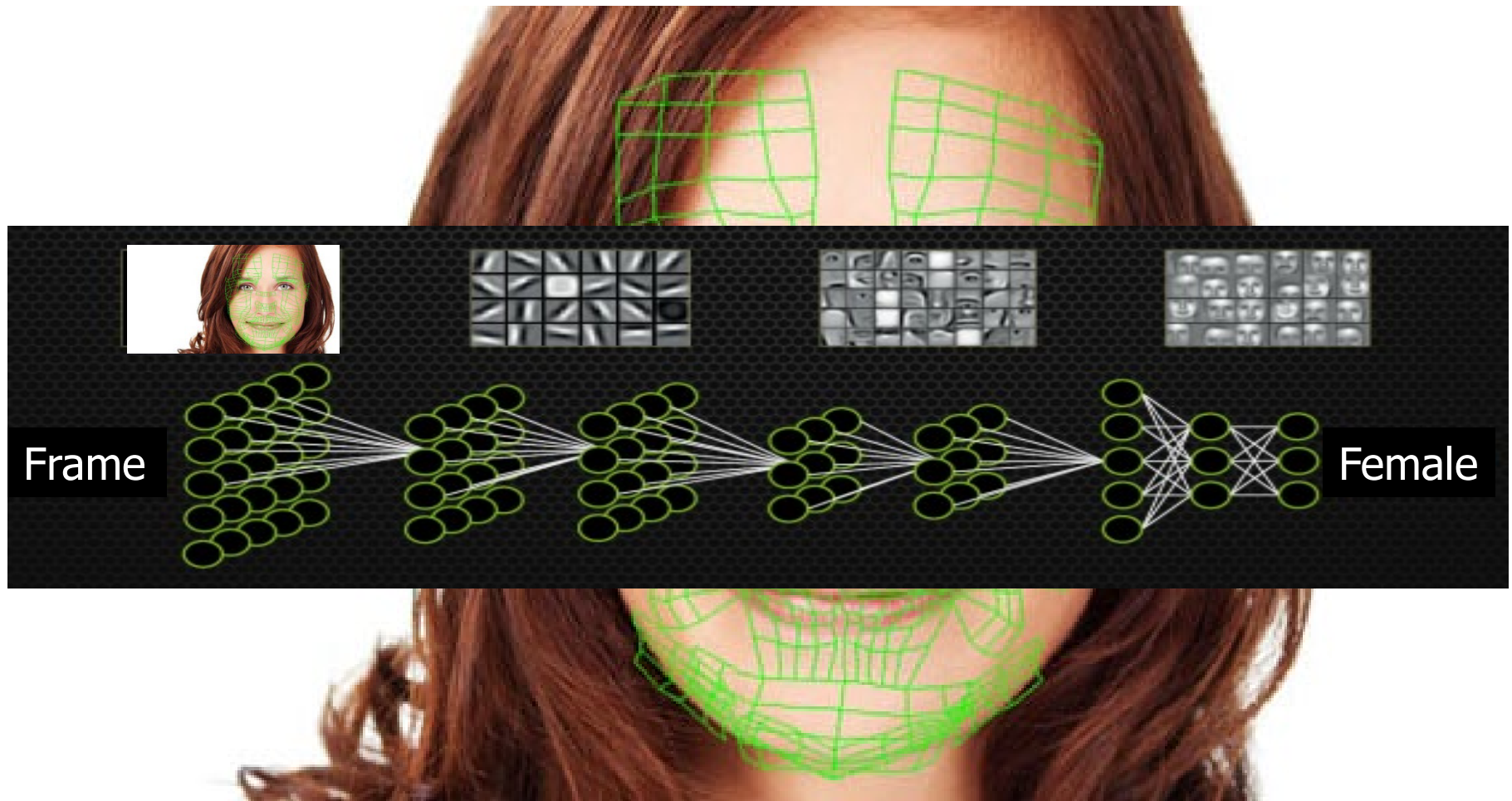
$$f(\mathbf{x}) = \text{sign}(\mathbf{w}^T \mathbf{x} + b)$$

Maximum Margin Classification: SVM

- Maximizing the margin is good.
- Implies that only support vectors matter; other training examples are ignorable.



Deep Learning: Convolutional Neural Networks



Large pretrained model



Sprouts in the shape of text 'Imagen' coming out of a fairytale book.



A photo of a Shiba Inu dog with a backpack riding a bike. It is wearing sunglasses and a beach hat.



A high contrast portrait of a very happy fuzzy panda dressed as a chef in a high end kitchen making dough. There is a painting of flowers on the wall behind him.



Teddy bears swimming at the Olympics 400m Butterfly event.



A cute corgi lives in a house made out of sushi.



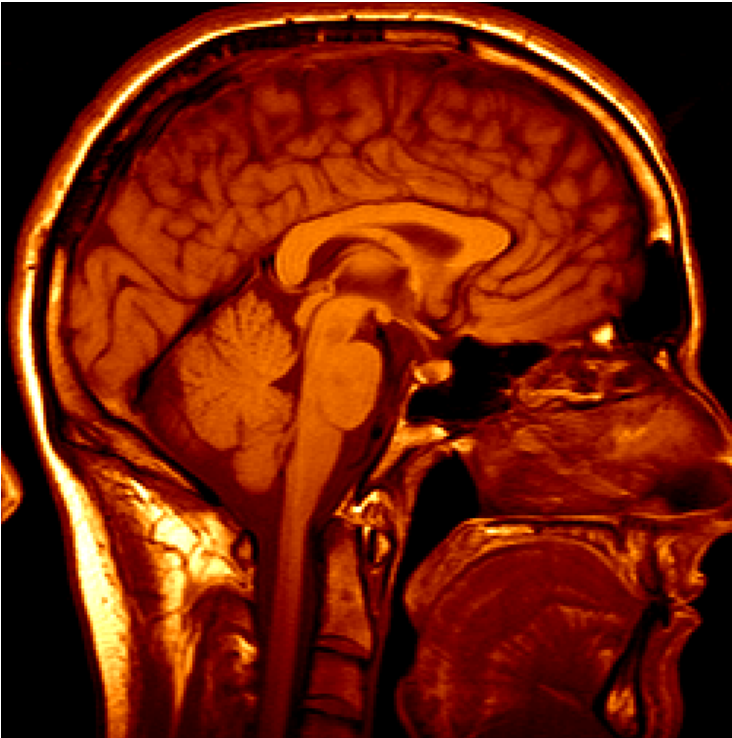
A cute sloth holding a small treasure chest. A bright golden glow is coming from the chest.

Computer Vision:
many applications

Object Tracking and autonomous driving



Medical imaging



3D imaging
MRI, CT



Image guided surgery
[Grimson et al., MIT](#)

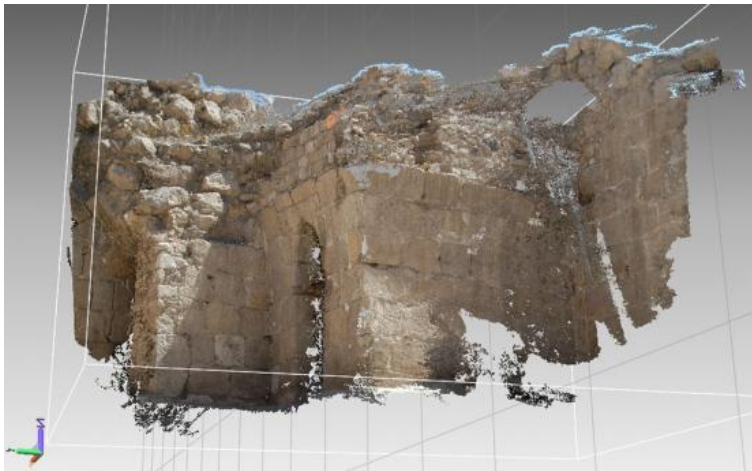
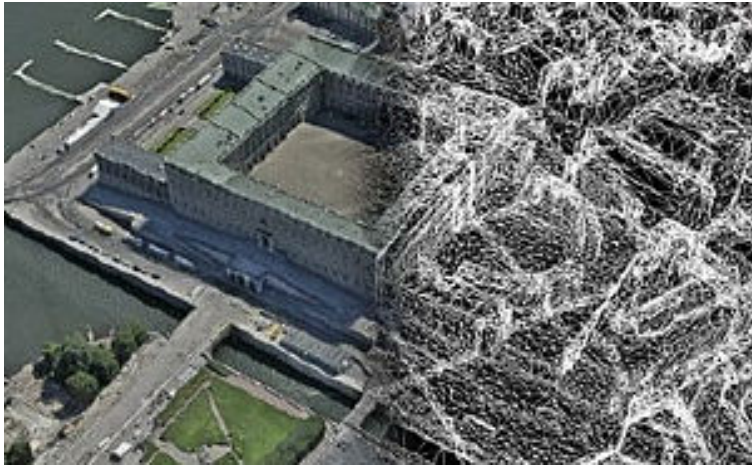
Augmented reality, consumer products



Special effects: shape and motion capture



3D from Images



Building Rome in a Day: Agarwal et al. 2009

Vision for robotics, space exploration

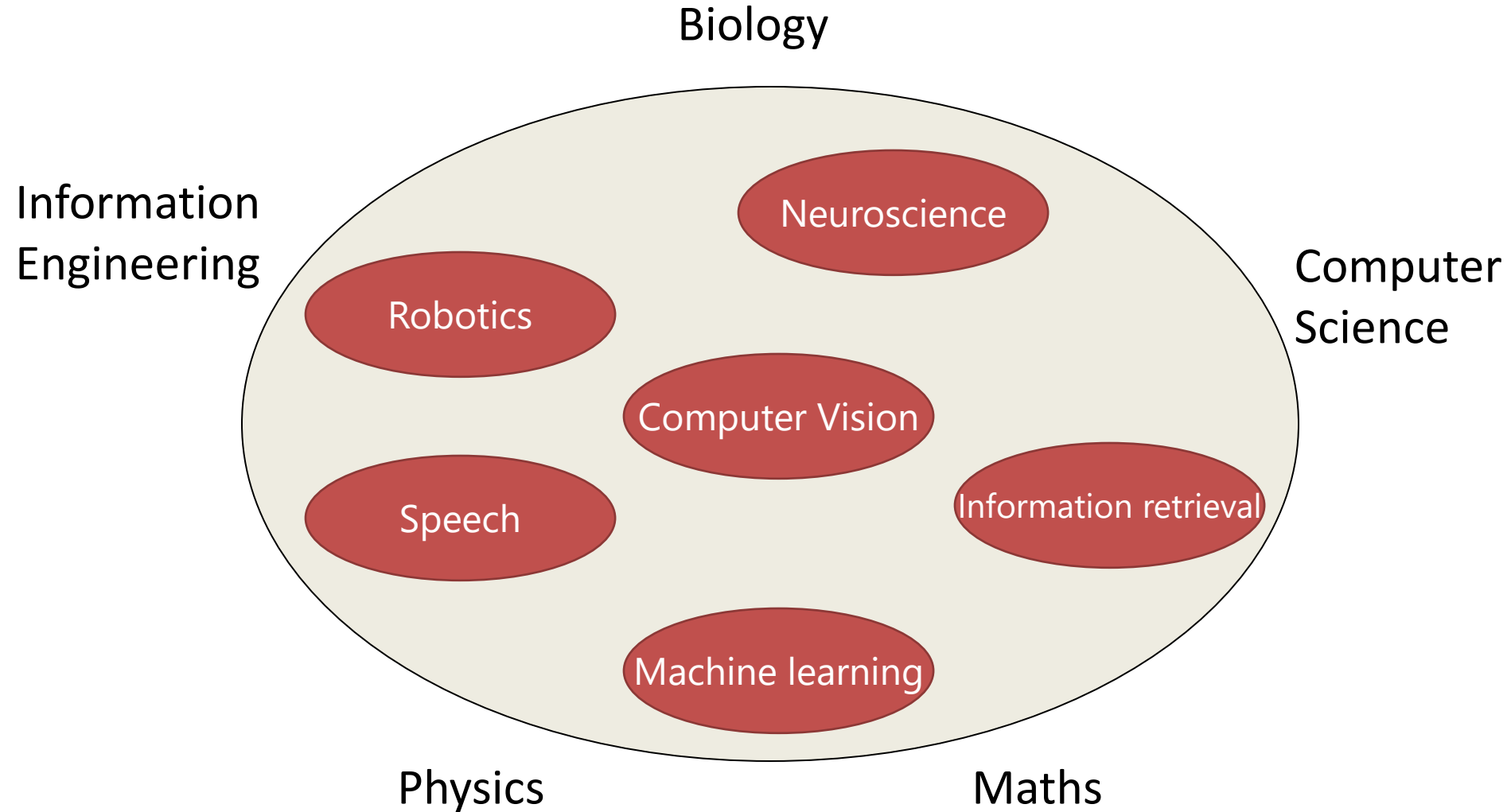


[NASA'S Mars Exploration Rover Spirit](#) captured this westward view from atop a low plateau where Spirit spent the closing months of 2007.

Vision systems (JPL) used for several tasks

- Panorama stitching
- 3D terrain modeling
- Obstacle detection, position tracking

Computer vision is a multidisciplinary research field



The programme is completed with the Master Thesis AI (36 EC) starting in the second period of the second year of the Master.

	Sem. 1	Sem. 2	EC
➤ Computer Vision 1	  	  	6
➤ Machine Learning 1	  	  	6
➤ Natural Language Processing 1	  	  	6
➤ Deep Learning 1	  	  	6
➤ Fairness, Accountability, Confidentiality and Transparency in AI	  	  	6
➤ Information Retrieval 1	  	  	6
➤ Reinforcement Learning	  	  	6
➤ Master Thesis AI	  	  	36
➤ <i>Constrained choice</i>	  	  	30
➤ <i>Suggested elective courses</i>	  	  	0

Administrative information

- Administrative information
- What you will expect in this course.

Teaching Staff



Dr. Martin Oswald
Coördinator
Lecturer



Dr. Arnoud Visser
Lecturer



Qi Bi
Senior Teaching Assistant

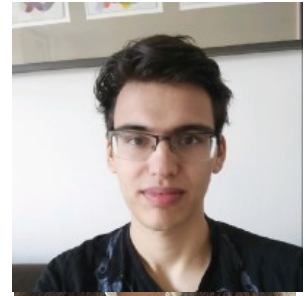
TA's



Vladimir Yugay



Roan van Blanken



Owen de Jong



Marina O.G.



Oliver Neut



Adrian Sauter



Matey Krastev



Konrad Szewczyk



Egoitz Gonzalez



Floris Six Dijkstra



Ronny Sandoval



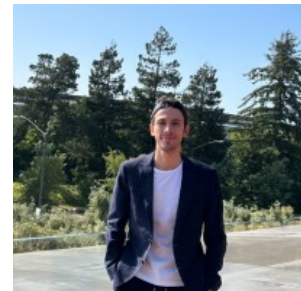
Yue Li



Melis Öcal



Xiaoyan Xing



Tony Tragoudaras

Important information of the course

- An introductory-level course for computer vision.
- Designed for 1st-year postgraduate students, particularly master of AI students.
- It contains: 14 sessions of 2-hour lectures, 5 computer labs (Python-based), 5 theoretical exercises and a final exam.

Assessment components

	Lab assignments	Exercises	Exam	Bonus
Total	5 labs in total 50%	5 %	45%	Up to 5%
Remarks	Teamwork, 3 students each team	5 canvas quiz, each worth 1%	45pts must get at least 25pts to pass	For the final projects

Course outline

Check regularly on Canvas->Modules->General Information

Week	Lecture 1 Monday 1-3pm C1.110	Lecture 2 Thursday 3-5pm C1.110 / L1.02	Textbook and other study material	Seminar Tuesday 11am-1pm L1.02	Lab Check you group info
1 2 - 6 Sep	Introduction and Course Logistics Camera model Lecturer: Arnoud Visser	Sensor and image formation Color Lecturer: Arnoud Visser	Computer Vision: 1.1, 1.2; 2.1.1, 2.1.2, 2.1.4, 2.2, 2.3.1, 2.3.2	No seminar this week Self check on math	Python/Jupyter Notebook installation tech support Introductory: •Python for image processing •Numpy
2 9 - 13 Sep	In place image processing Photometric stereo Lecturer: Arnoud Visser	Convolution and image filtering Lecturer: Arnoud Visser	Computer Vision: 3.1.1, 3.1.2, 3.1.4, 13.1.1; 3.2; 3.3.	Tutorial 1: Image formation and Image processing Questions give out (Monday 9/9) Teacher: Qi Bi	Lab 1: Photometric Stereo and Color Assignment give out (Friday 6/9) Submission deadline (23:59 Saturday 14/9)*
3 16 - 20 Sep	Feature: edge and corner Lecturer: Arnoud Visser	Motion and optical flow Lecturer: Arnoud Visser	Computer Vision: 7.2, 7.4, 7.1; 9.1.1, 9.1.3, 9.3 (9.3.1 optional);	Tutorial 2: Convolution, Image filtering: Questions give out (Monday 16/9) Teacher: Qi Bi	Lab 2: Edge detection, convolution and image filtering Assignment give out (Friday 13/9) Submission deadline (23:59 Saturday 21/9)
4 23 - 27 Sep	Global transform, Image stitching, alignment Lecturer: Martin Oswald	Basics of Recognition 1 Lecturer: Martin Oswald	Computer Vision: 2.1, 8.1, 8.2; 13.1; 6.1, 6.2.1,	Tutorial 3: Local Feature and Motion Questions give out (Monday 23/9) Teacher: Qi Bi	Lab3: Optical flow and motion segmentation Assignment give out (Friday 20/9) Submission date (23:59 Saturday 28/9)
5 30 - 4 Oct	Basics of Recognition 2 Lecturer: Martin (online)	Convolutional neural network 1 (Basic) Lecturer: Martin (online)	Computer Vision: 5.2.1, 5.2.2; 6.3, 6.4(introduction), 6.4.2	Tutorial 4: BoW, Image classification Questions give out (Monday 30/9) Teacher: Qi Bi	Lab 4: Feature extraction, Image alignment and segmentation Lab4 Assignment give out (Friday 27/9) Submission deadline (Saturday 5/10)
6 7 - 11 Oct	Convolutional neural network (Training and testing) Lecturer: Martin (online)	Stereo, 3D reconstruction Lecturer: Martin Oswald	Computer Vision: 5.3, 5.4; 12.1, 12.2, 12.3, 13.2.	Tutorial 5: CNN Question give out (Monday 7/10). Teacher: Qi Bi	Final Lab: Part 1: Bag of Words for image classification Lab Assignment give out (Friday 4/10) Submission deadline (Saturday 19/10)
7 13 - 18 Oct	Application and guest lecture 1 TBA Application and guest lecture 2: TBA	Application and guest lecturer 3 TBA Course Wrap-up Lecturer: Martin Oswald		Tutorial 6 Sample exams Question and answers for exam. Teacher: Qi Bi	Final Lab: Part 2: CNN for object recognition Lab Assignment give out (Friday 4/10) Submission deadline (Saturday 19/10)

Content

Content of This Course	Compare with UvA's Bachelor Course
Camera model	+
Sensor and Image formation	New
Homogenous coordinates	+
Color	New
In place image processing	Mostly New
Photometric stereo and shape from X	New
Convolution and image filtering:	+
Feature: edge and corner	+
Motion	+
Optical flow	Mostly New
Global transform, Image stitching, alignment	+
Bag of Words	New
Image retrieval, image classification, Object detection	New
Convolutional neural network	Mostly New
Stereo, 3D reconstruction	New

Lectures

- 14 Lectures in total, which covers theories and knowledge about computer vision.
- This is master course; the lecture will not cover everything. Self study (slides, textbook and search online) is important!
- The lecture is physical, recordings will be published afterwards.

Lectures:

Intended Learning Outcome

- Understand the base concept (What) and motivation (Why) of computer vision
- Understand the key theories and techniques (How)
- Know the trend of modern computer vision.
- Gain the basic knowledge for future courses, such as:
 - Computer Vision 2
 - Deep Learning

Exercises (Seminar)

Logistics

- Week2 -6, every week a quick exercises will be published (canvas), which takes 5-10mins. Graded
- Every Tuesday 11am-1pm we host a Seminar at L1.02 for practicing, solving and Q&A on example questions for the exam.
- Week 7, we host a Seminar at L1.02 for Q&A on final exam. Sample exams will be released before that.

Exercises (Seminar)

Graded

- Exercises are example questions that help you to understand the theory and prepare for the final exam.
- 5 graded exercises (1% each; 5% total) will be shared on Canvas each week on Monday and **MUST** be finished before the seminar.
- The last seminar will discuss example questions for the exam.
- Where is it:
 - Quiz: Canvas->Assignments
 - Exam style questions: Canvas -> Modules -> Exercises Week X

Exercises (Seminar)

Intended Learning Outcome

- Practice and use the key theories and techniques, with an emphasis on the theory.
- Can reflect on the pros and cons of existing algorithms.
- Be prepared for future courses.

Lab assignments

Canvas -> Modules -> Week X

- Assignments with detailed instruction will be given out on Canvas.
- Labs 1 – 5 are graded, in total 50%.
 - Labs 1-4 weights: 12%, 14%, 14%, 10%
 - Final lab weight: 50%
- Teamwork, each team has three students. Each team only need to submit once to Canvas. We will pre-allocate teams for you.
- If you have reasons to reassign the team, please contact Senior TA Qi Bi (q.bi@uva.nl)
- Coding: Python
- Feel free to ask questions on piazza.
- Quality demands in Course Manual

Labs: Intended Learning Outcome

- Practice and use the key theories and techniques which an emphasis on practice.
- Can reflect on the pros and cons of existing algorithms.
- Can propose improvement from existing methods.
- Be prepared for future courses.

Labs:

Logistics for physical sessions

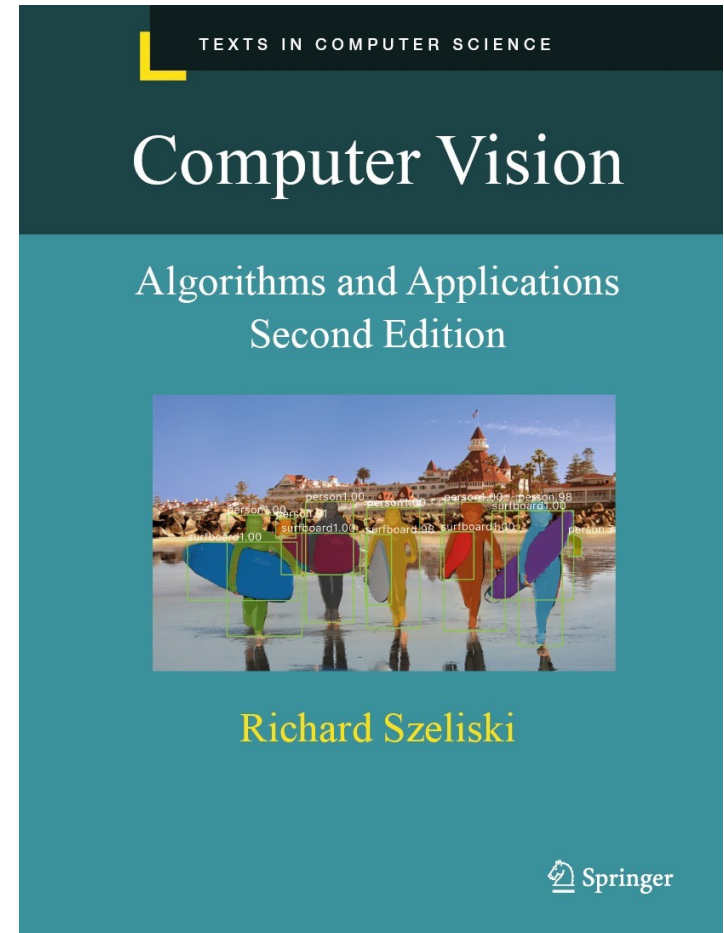
- Every Thursday we host lab-office hours (also in week 1).
 - 5 groups on Thursday, 3 groups on Friday
- We may change the schedule, keep an eye on canvas (canvas is leading over datanose) !!
- Not mandatory
- You are assigned to a group (datanose).
- If room space allows, you may choose any of time/rooms available.

Final Exam (45%)

- Physical, Passenger Terminal *
- Must be on your own.
- 13:00-16:00 21 Oct 2024, Monday
- *time and place, check announcement regularly
- Sample exams will be released around Week 6.

Textbooks

- CV1 is based on “Computer Vision: Algorithms and Applications” by Richard Szeliski
 - Free for download from <http://szeliski.org/Book/>
 - Use the 2022 version

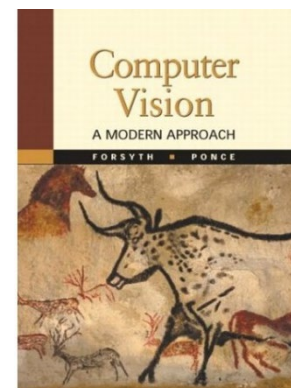
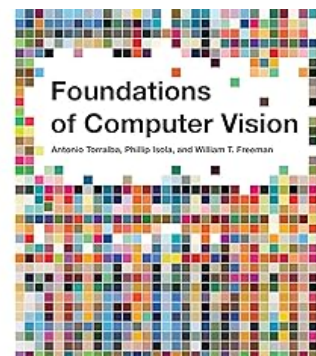
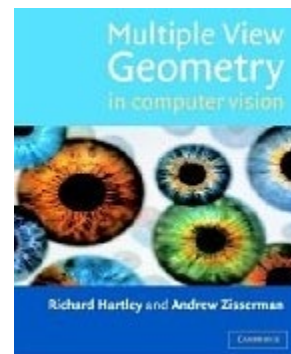


Prerequisites

- Canvas -> Modules -> Week0 -> Before you start
- Linear algebra, calculus, and probability
 - Check Canvas->Module->Math Revision for detail.
- Machine learning (in parallel)
- Data structures
- Python

Three other useful books

- Hartley, R. and Zisserman, A. Multiple View Geometry in Computer Vision, Academic Press, 2002.
- Forsyth, David A., and Ponce, J. Computer Vision: A Modern Approach, Prentice Hall, 2003.
- Antonio Torralba, Phillip Isola, William T. Freeman, Foundations of Computer Vision, MIT Press, 2024.

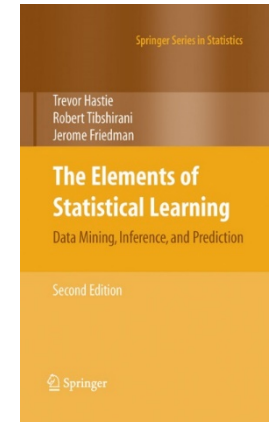
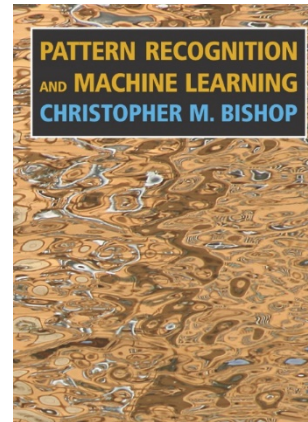


Even More Background and Tools

- Machine Learning books:

Bishop, C. Pattern Recognition and Machine Learning, 2006.

Hastie, T., Tibshirani R., and Friedman J,
The Elements of Statistical Learning,
Freely available for download:
<http://www-stat.stanford.edu/~tibs/ElemStatLearn/>



- Software, programming tools

Matlab, image processing toolbox,
C++, openCV
Python, PIL, numpy, scipy
PRTTools, Shogun, (machine learning
toolboxes)



- Scientific papers:

Top Journals: Transactions of Pattern Analysis and Machine Intelligence, International Journal of Computer Vision, TIP, (Computer Vision and Image Understanding)

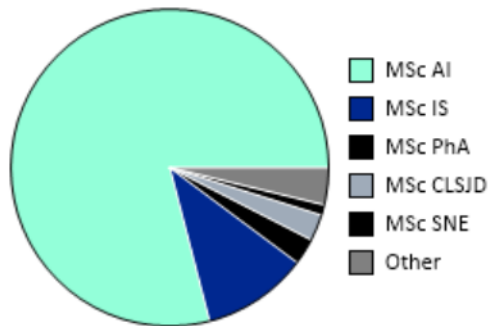
Top Conferences: ICCV, ECCV, CVPR, NIPS, SIGGRAPH, BMVC, (ACM-Multimedia)

<http://www.cvpapers.com/>

Some statistics

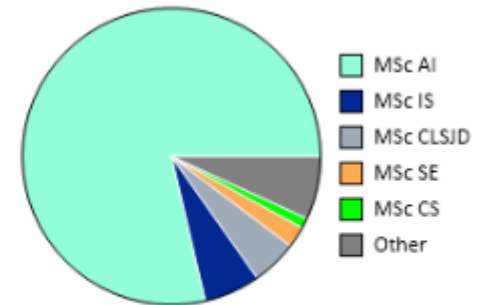
Program of students

2022/2023 ▼



First attempt: 94% (139)

2023/2024 ▼

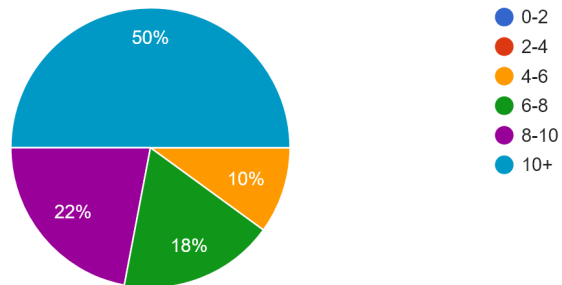


First attempt: 90% (186)

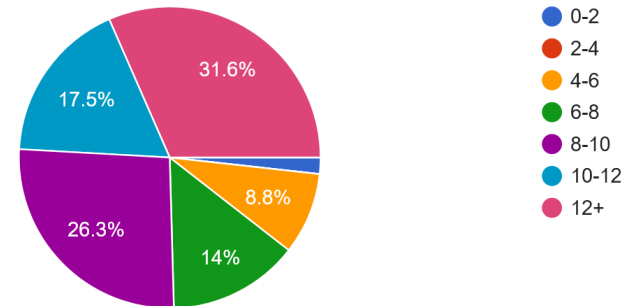
Working load

How many hours you spend on each assignment

50 responses



2021/2022



2022/2023

In 2022 ML1 reduced some working loads 😊

Action Points

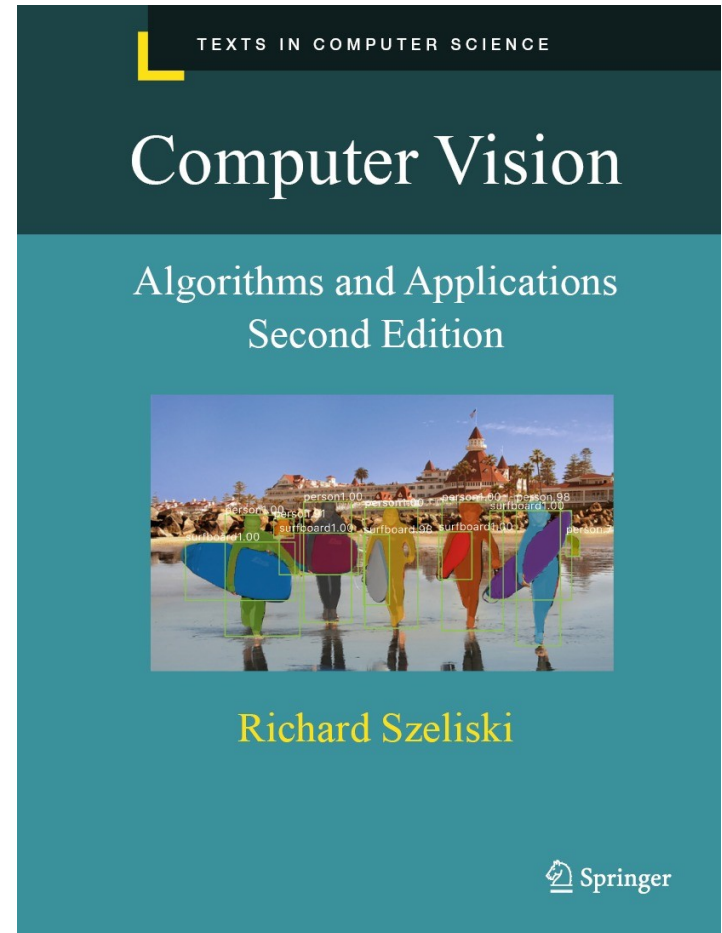
- Check information regularly on Canvas
- Download the textbooks
- Find your group members!
- Use the Math Revisions checklist
- Install python and practice numpy and basis image processing (exercise 1: lab0), go to lab-office hours if needed (Thursdays/Fridays)
- Find your time to attend physical sessions of seminars and labs (not mandatory)
- Piazza

Computer Vision 1

1-1: Camera Model and Image Formation

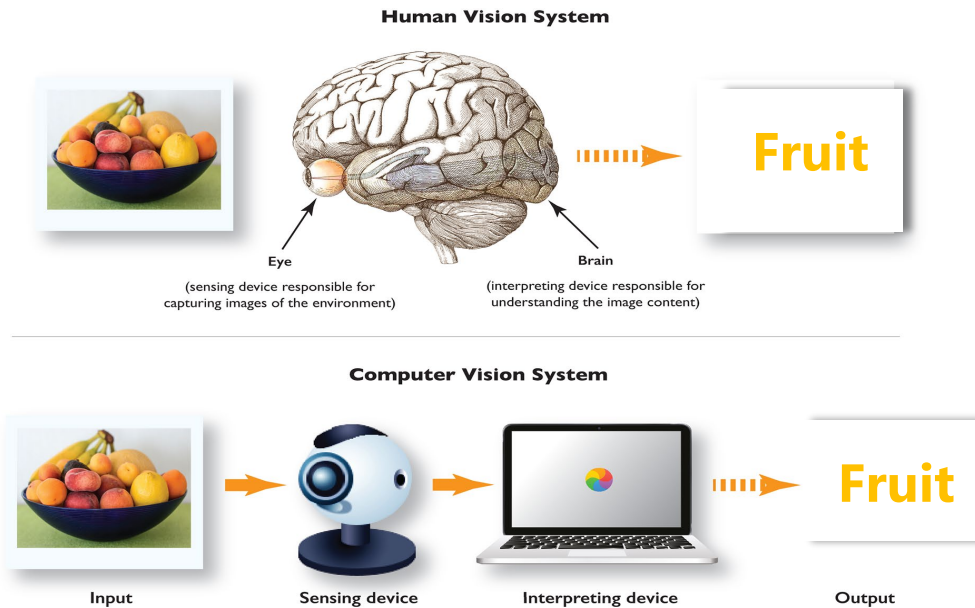
Textbook

- 2.1.1
- 2.1.2
- 2.1.4



The computer vision problem

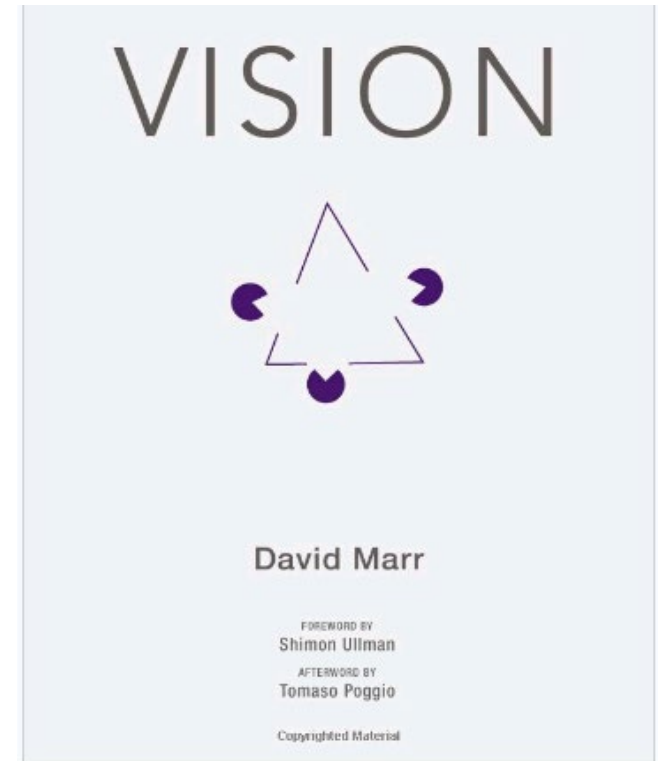
- Make a computer see and understand images



- We know it is physically possible –
we do it every day and effortlessly!

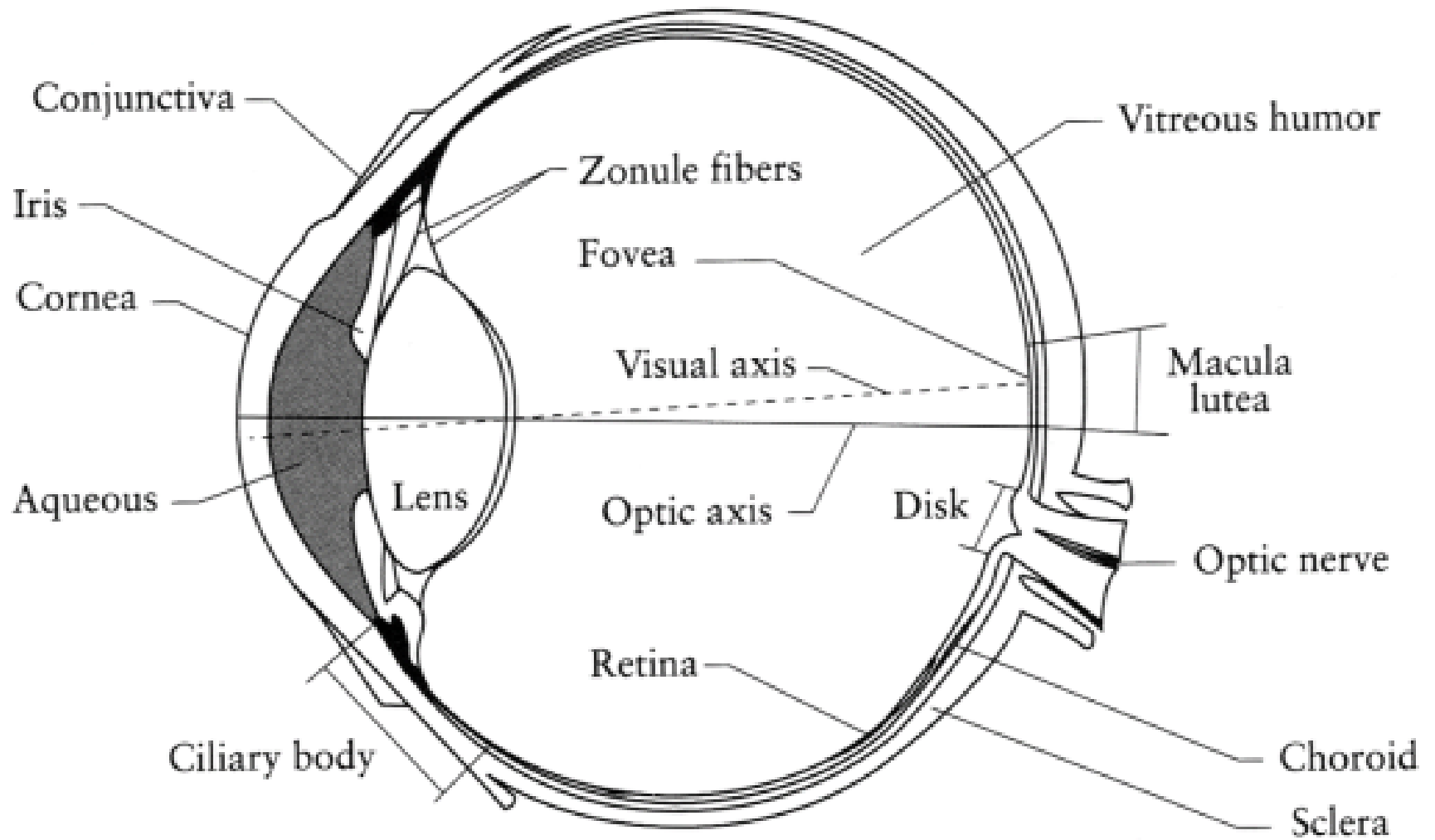
Three levels of vision processing

- Low-level vision:
 - image processing, denoising, filtering, image restoration, low-level feature extraction.
- Mid-level vision:
 - image analysis, image segmentation, contour extraction, perceptual organization, 2-1/2D representation, 3D information recovery.
- High-level vision:
 - visual detection, recognition, understanding, semantic labelling, activity, event detection.

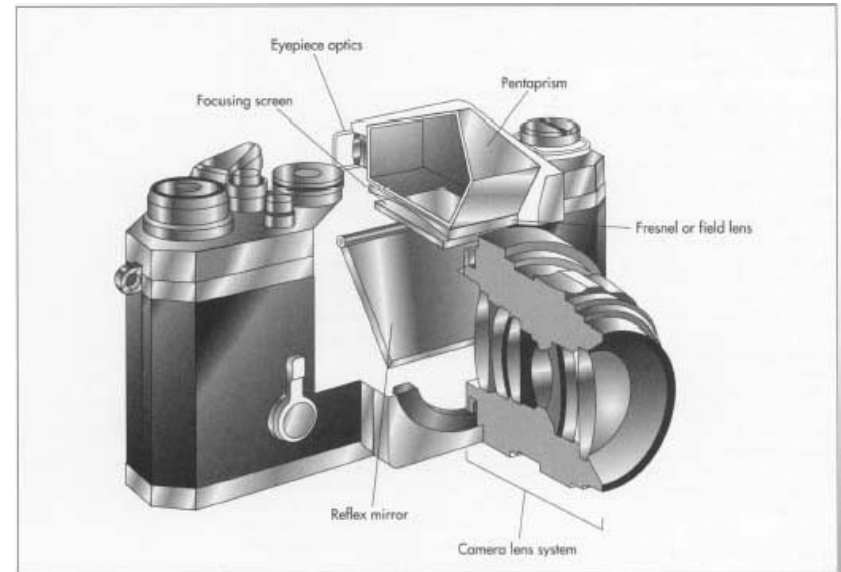


Outline

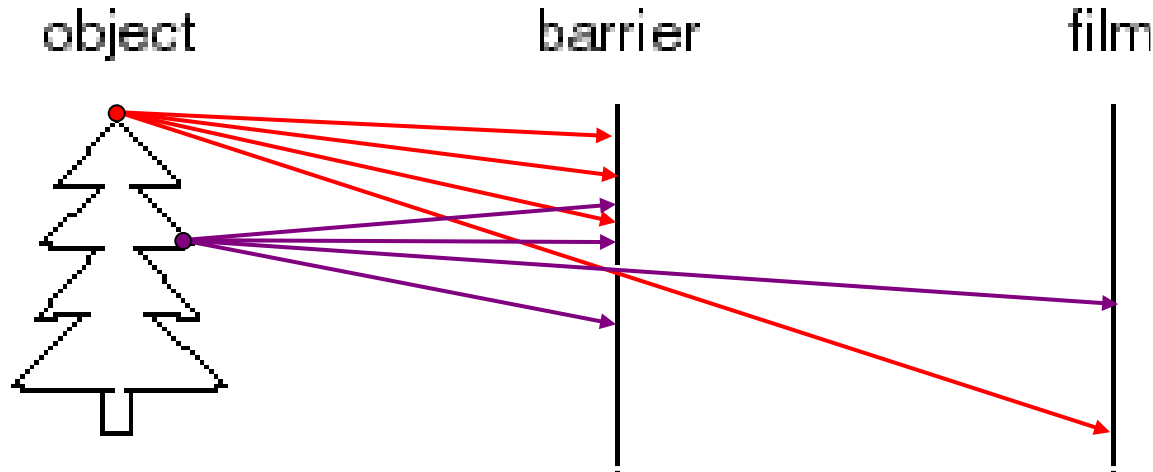
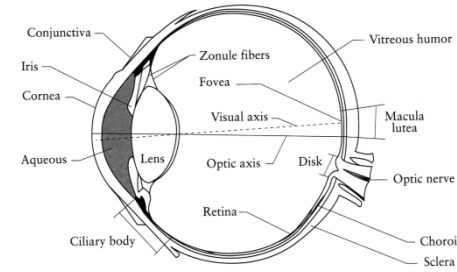
- A pinhole camera
- Geometric image formation
- Photometric image formation
- Image representation



What is a camera ?

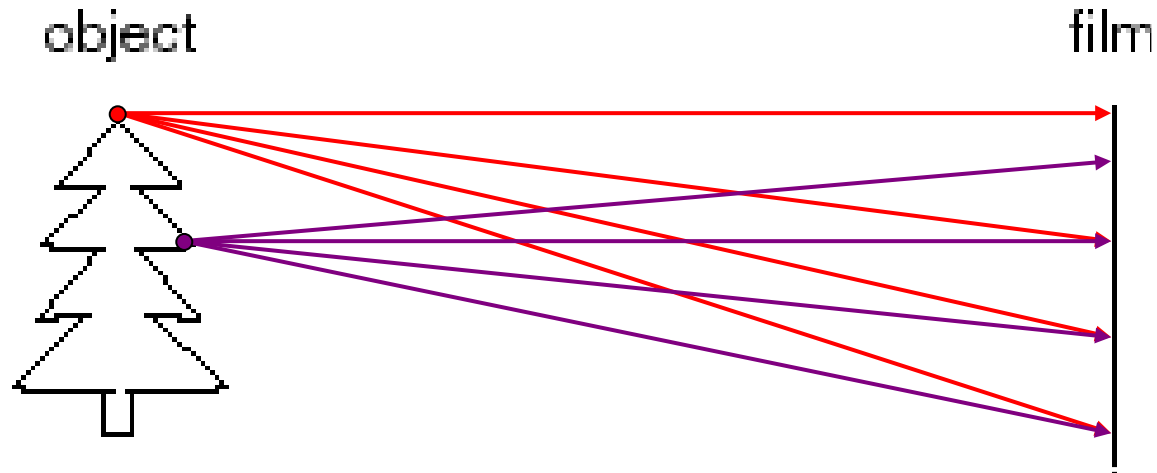


Pinhole camera



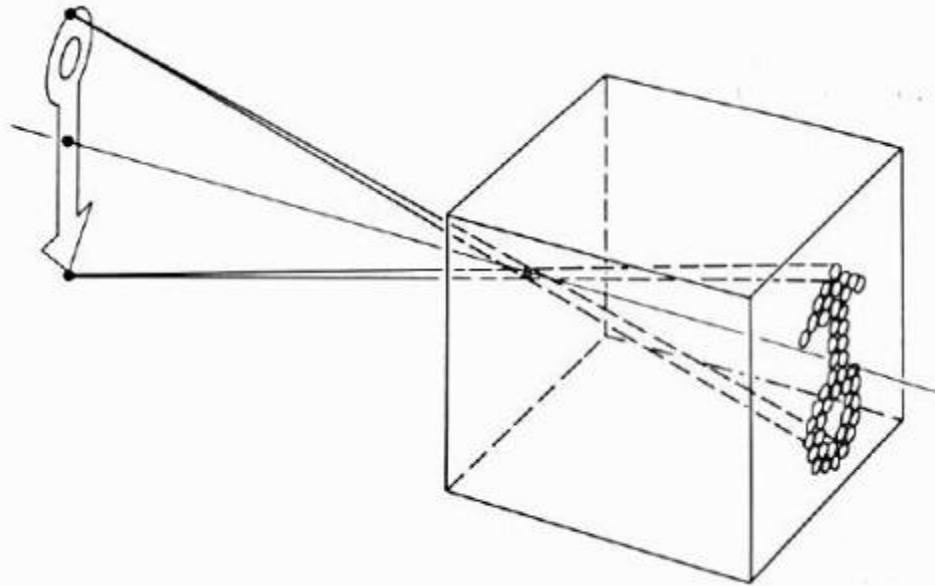
- Add a barrier to block off most of the rays
 - This reduces blurring
 - The opening is known as the **aperture**

Let's design a camera



- Idea 1: put a piece of film in front of an object
- Do we get an image?

Pinhole camera model



- Pinhole model:
 - Captures **pencil of rays** – all rays through a single point
 - The point is called **Center of Projection (focal point)**
 - The image is formed on the **Image Plane**

Make a camera obscura from your room

- Image is **inverted**
- **Depth of the room** is the **focal length**
- How can we **reduce the blur**?



How to build a camera obscura at home ([link](#))



Camera Obscura



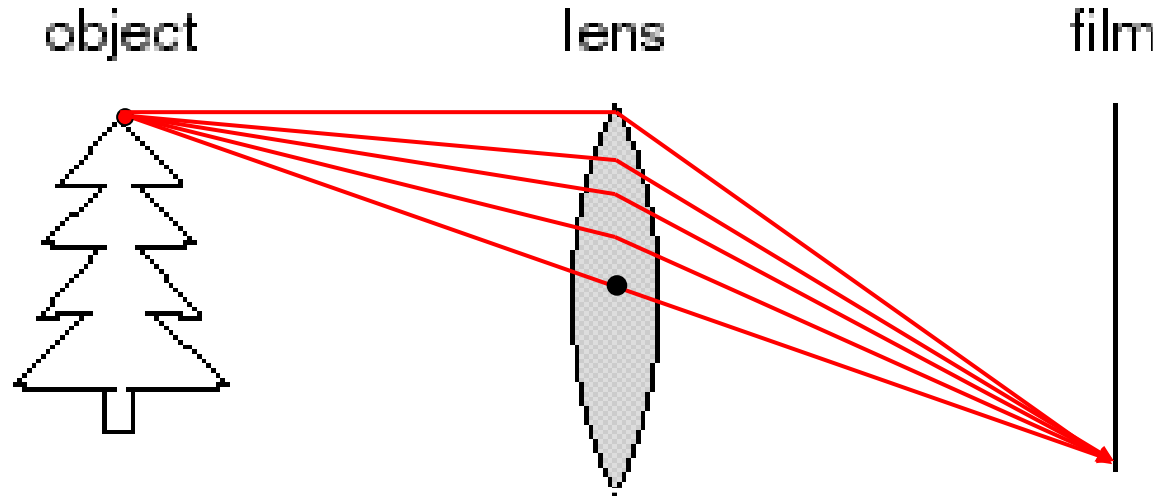
f/2.8



f/16

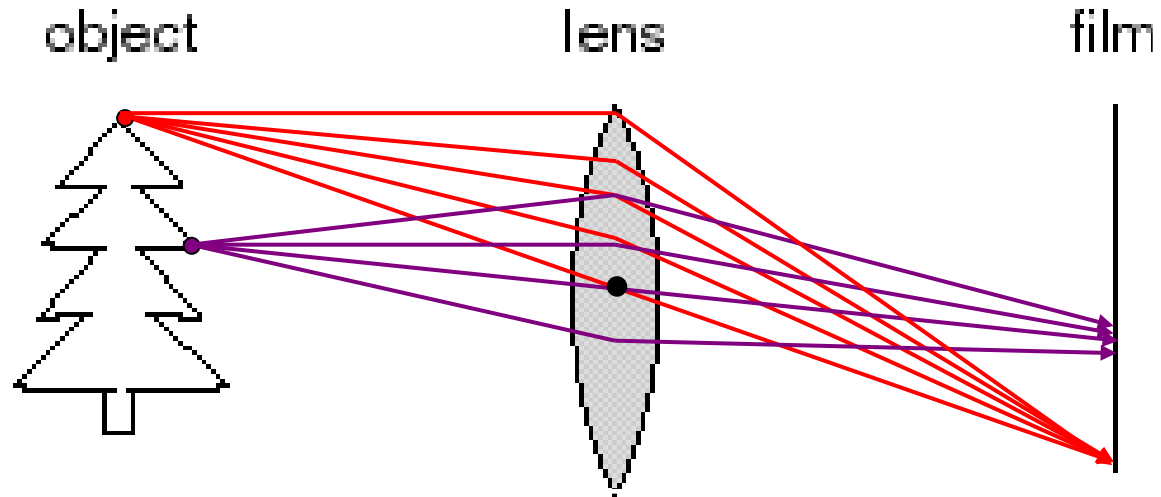


Adding a lens... to capture more light

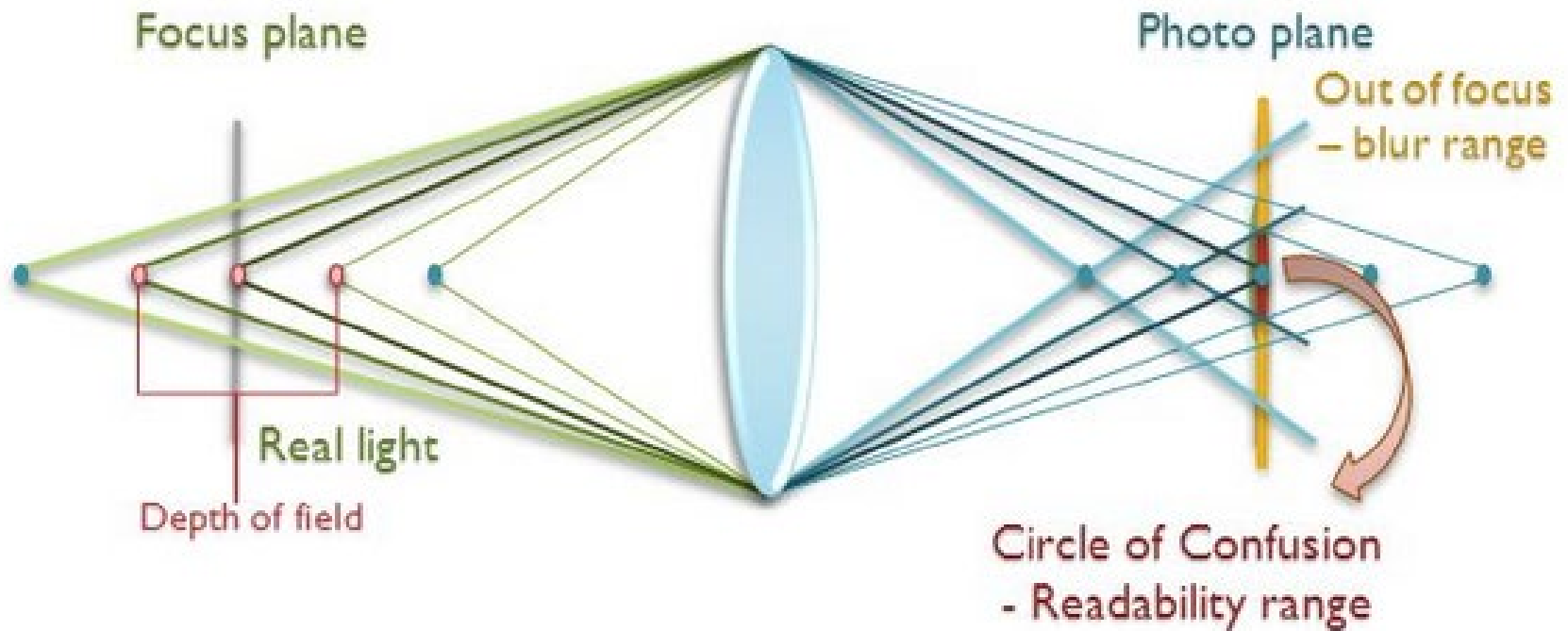


- A lens focuses light onto the film
 - Rays passing through the center are not deviated

Adding a lens



Depth and focus



Depth and focus

Aperture Adjustment Sequence - DOF



f/1.8



f/2.8



f/4.0



f/5.6

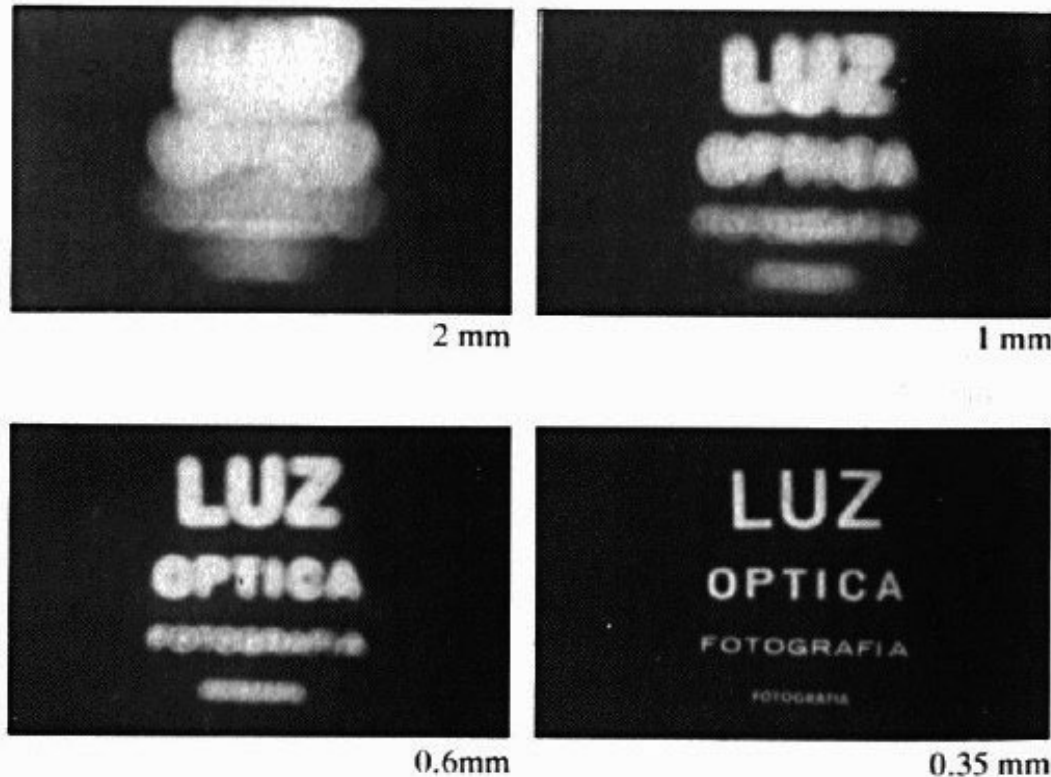


f/16



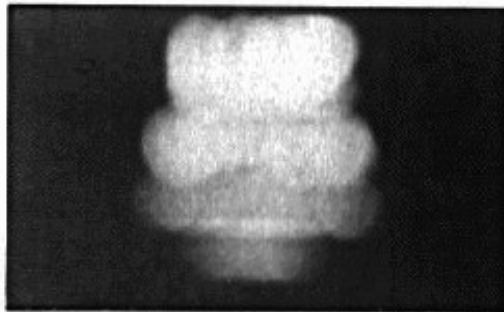
f/22

Shrinking the aperture

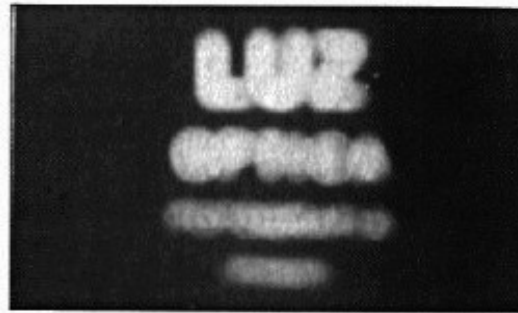


- Why not make the aperture as small as possible?
 - Less light gets through
 - Diffraction effects...

Shrinking the aperture



2 mm



1 mm



0.6mm



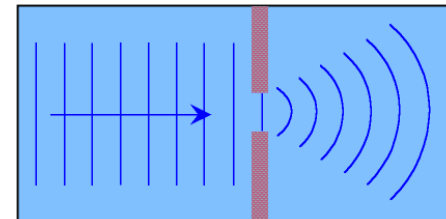
0.35 mm



0.15 mm



0.07 mm

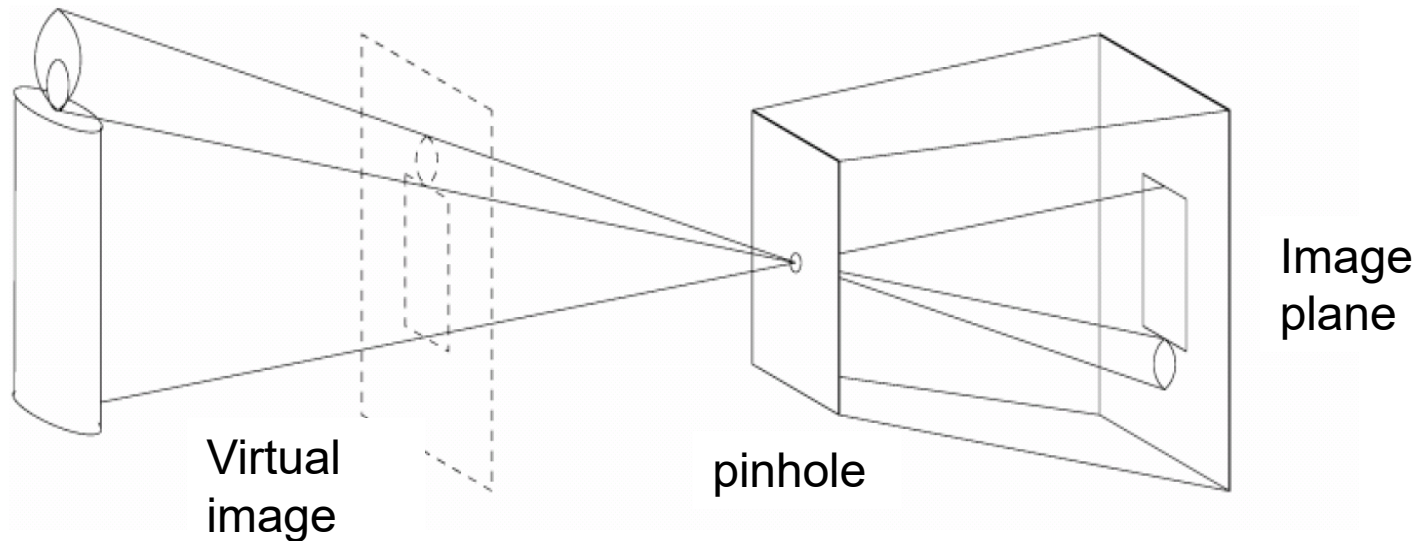


Outline

- A pinhole camera
- Geometric image formation
- Photometric image formation
- Image representation

Pinhole camera

- Pinhole camera is a simple model to approximate imaging process: perspective **projection**.



If we treat pinhole as a point, only one ray from any given point can enter the camera.

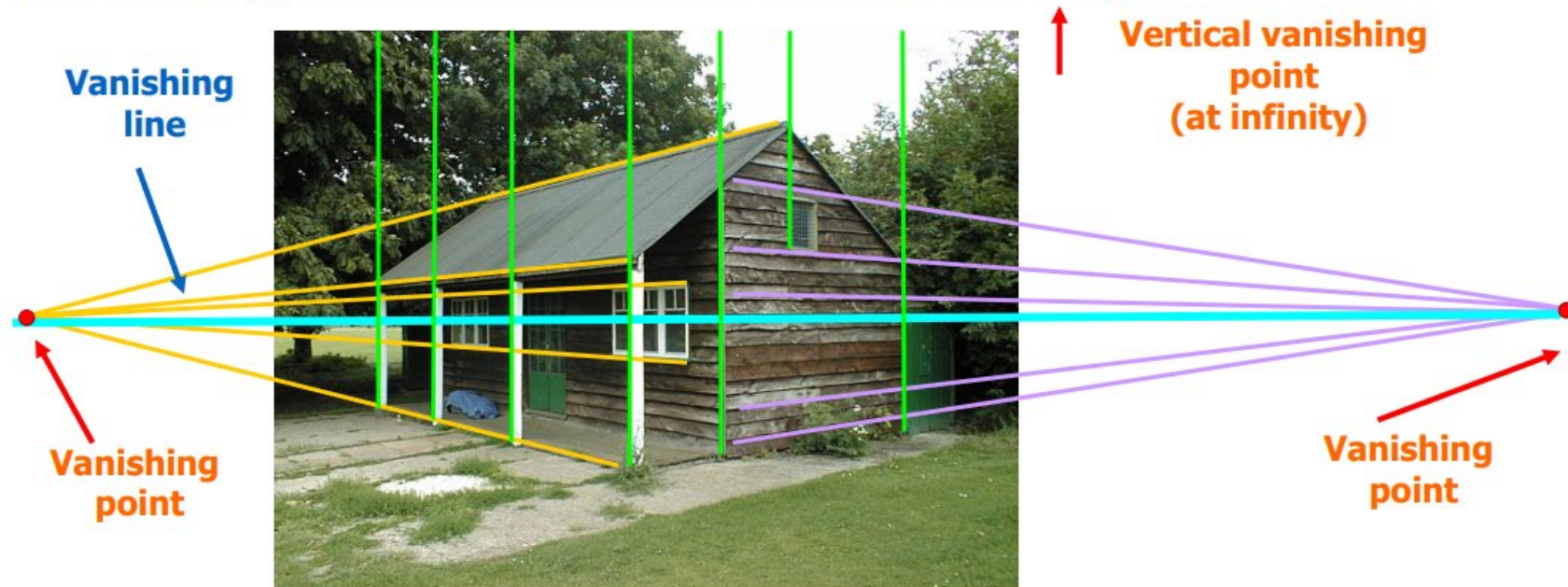
Perspective effects



3D computer vision aims to solve an inverse problem of computer graphics

Vanishing points and lines

- **Parallel lines** intersect at a “**vanishing point**” in the image
- **Parallel planes** intersect at a “**vanishing line**” in the image
- Notice that vanishing points can fall both inside or outside the image



Perspective in art

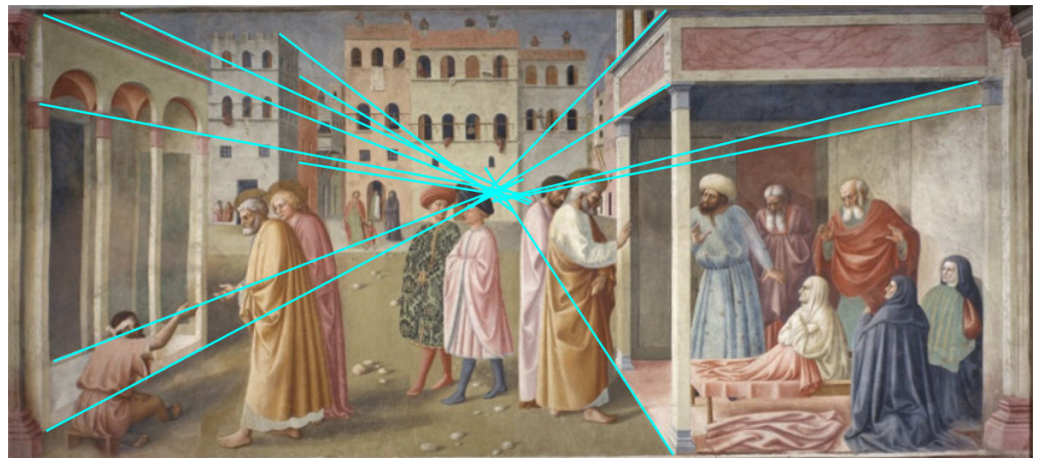
Oblique projection



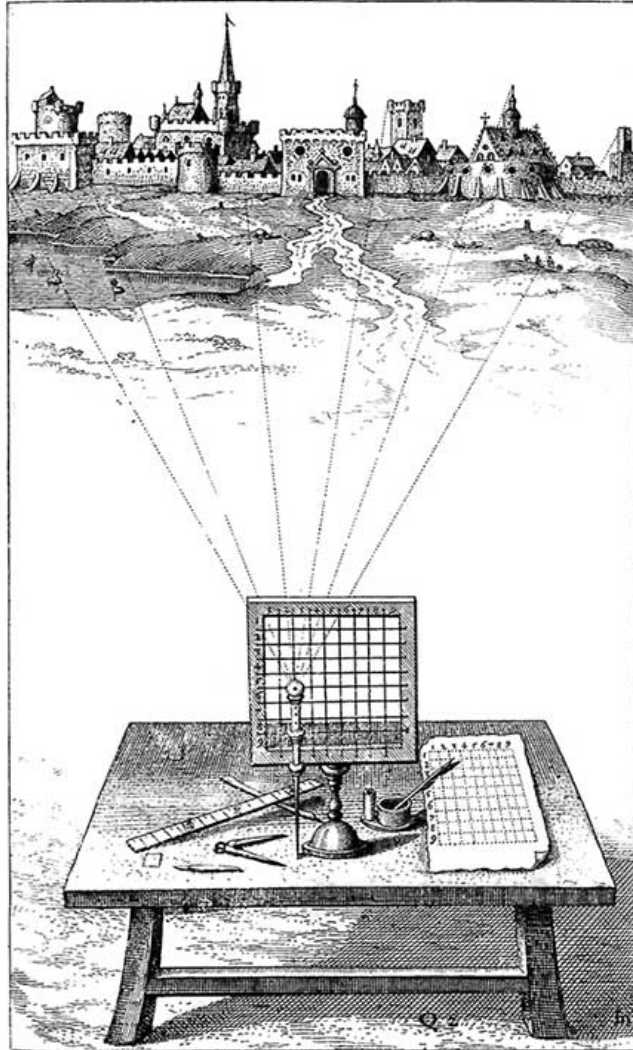
The Birth of Saint John the Baptist: Predella Panel
Giovanni di Paolo
1454

One-eye perspective

The Healing of the Cripple and Raising of Tabitha
Masolino
~1427
Cappella Brancacci, Florence



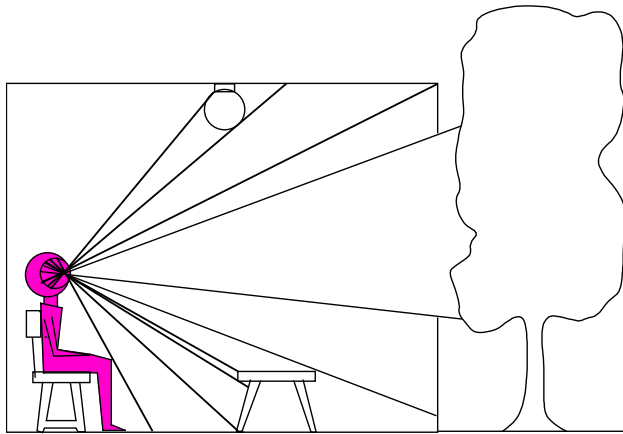
Perspective in art



Robert Fludd's [sighting grid](#) (1617)

Dimensionality Reduction Machine (3D to 2D)

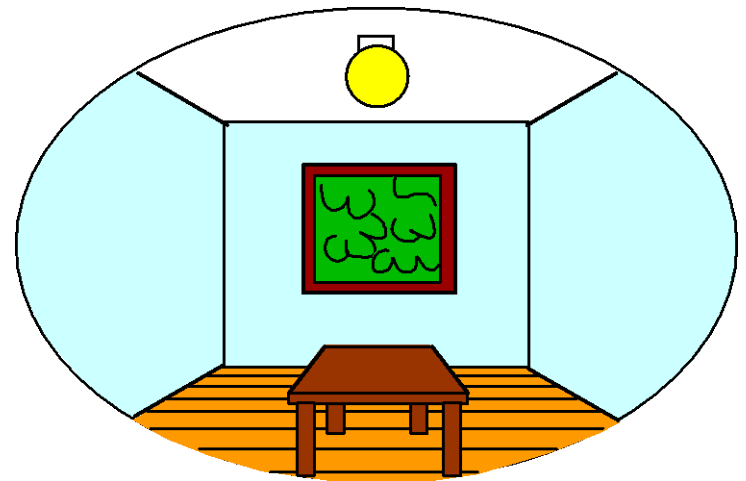
3D world



Point of observation



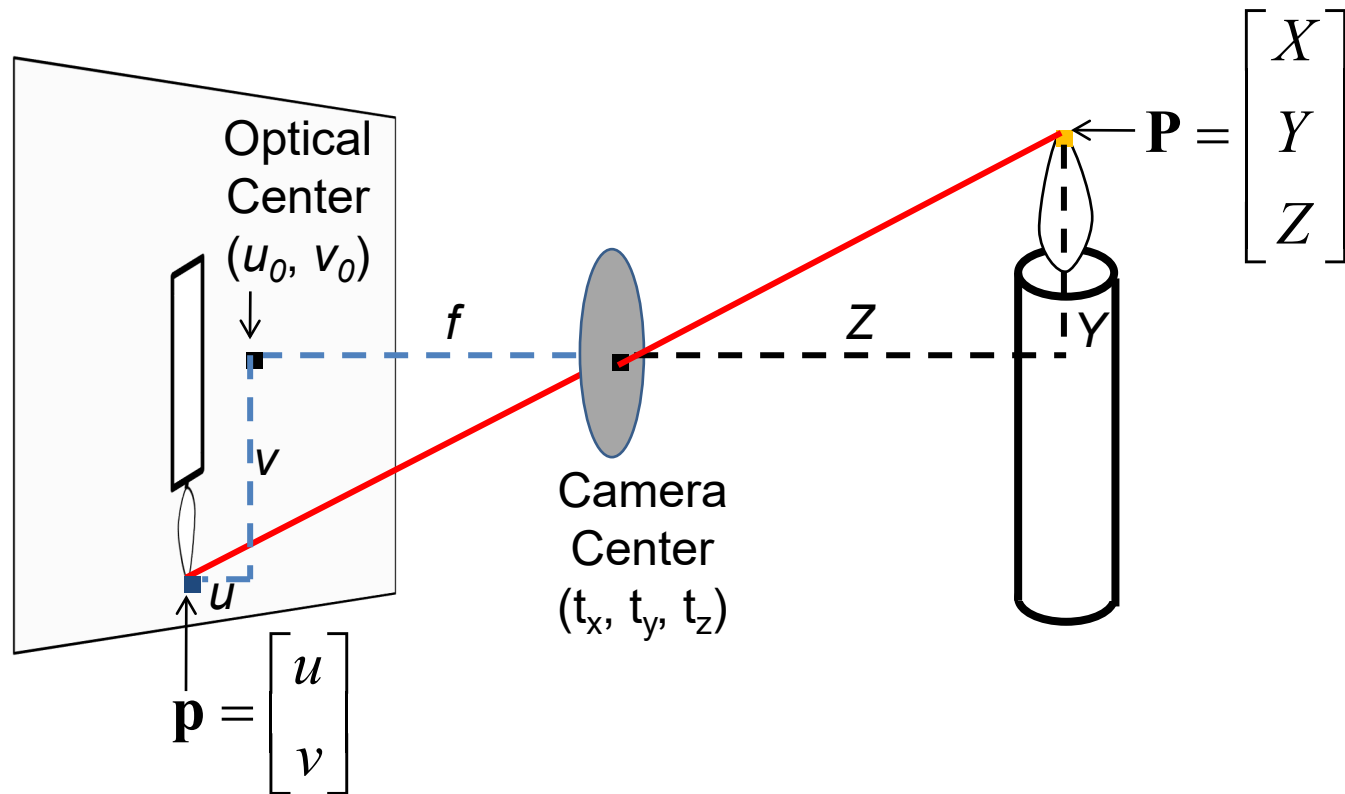
2D image



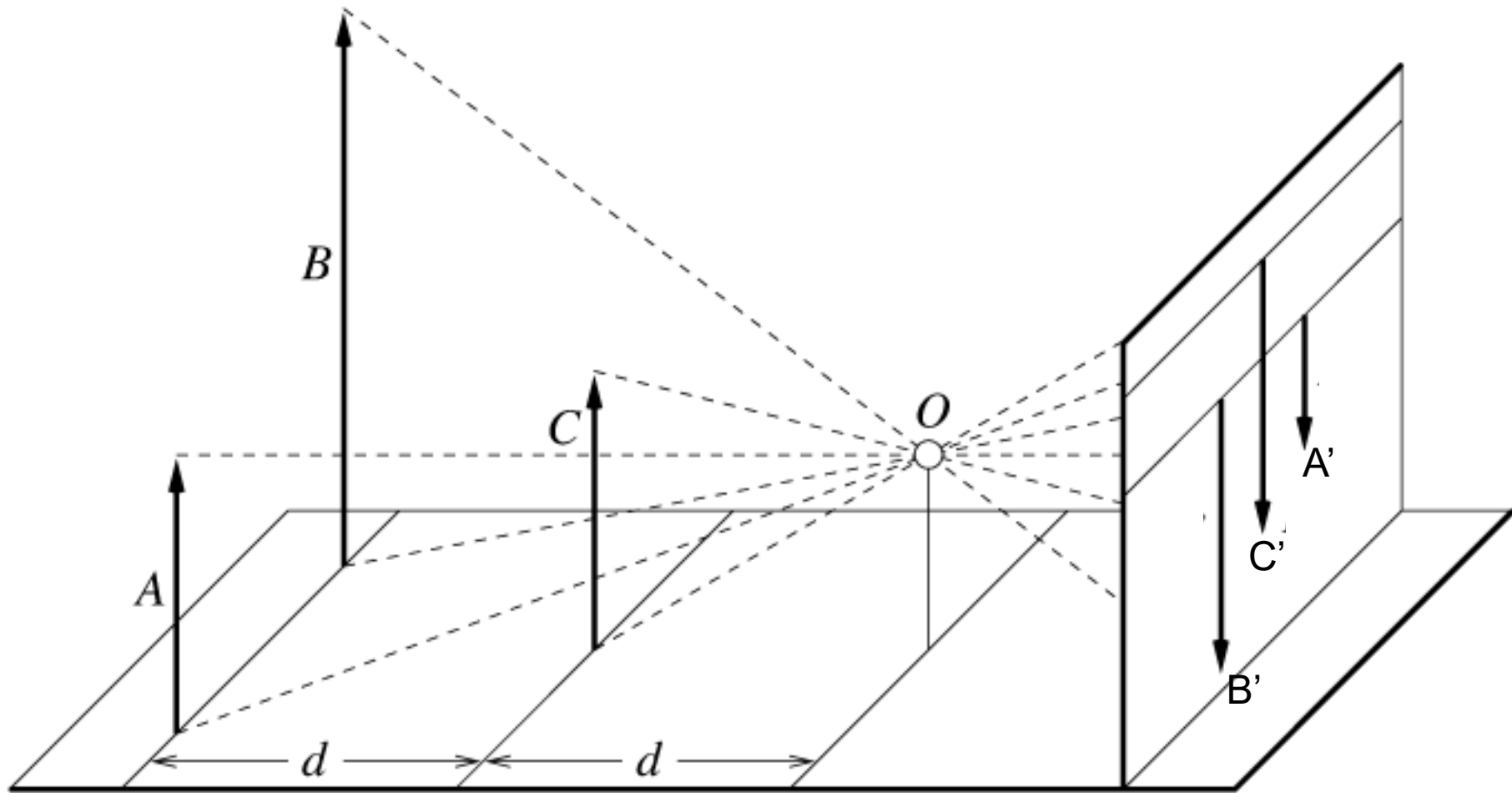




Projection: world coordinates to image coordinates



Length and area are not preserved



Outline

- A pinhole camera
- Geometric image formation
 - Homogeneous coordinates
- Photometric image formation
- Image representation

Homogeneous coordinates

Converting *to* homogeneous coordinates

$$(x, y) \Rightarrow \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

2D (image) coordinates

$$(x, y, z) \Rightarrow \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

3D (scene) coordinates

Converting *from* homogeneous coordinates

$$\begin{bmatrix} x \\ y \\ w \end{bmatrix} \Rightarrow (x/w, y/w)$$

2D (image) coordinates

$$\begin{bmatrix} x \\ y \\ z \\ w \end{bmatrix} \Rightarrow (x/w, y/w, z/w)$$

3D (scene) coordinates

Homogeneous coordinates

Invariant to scaling

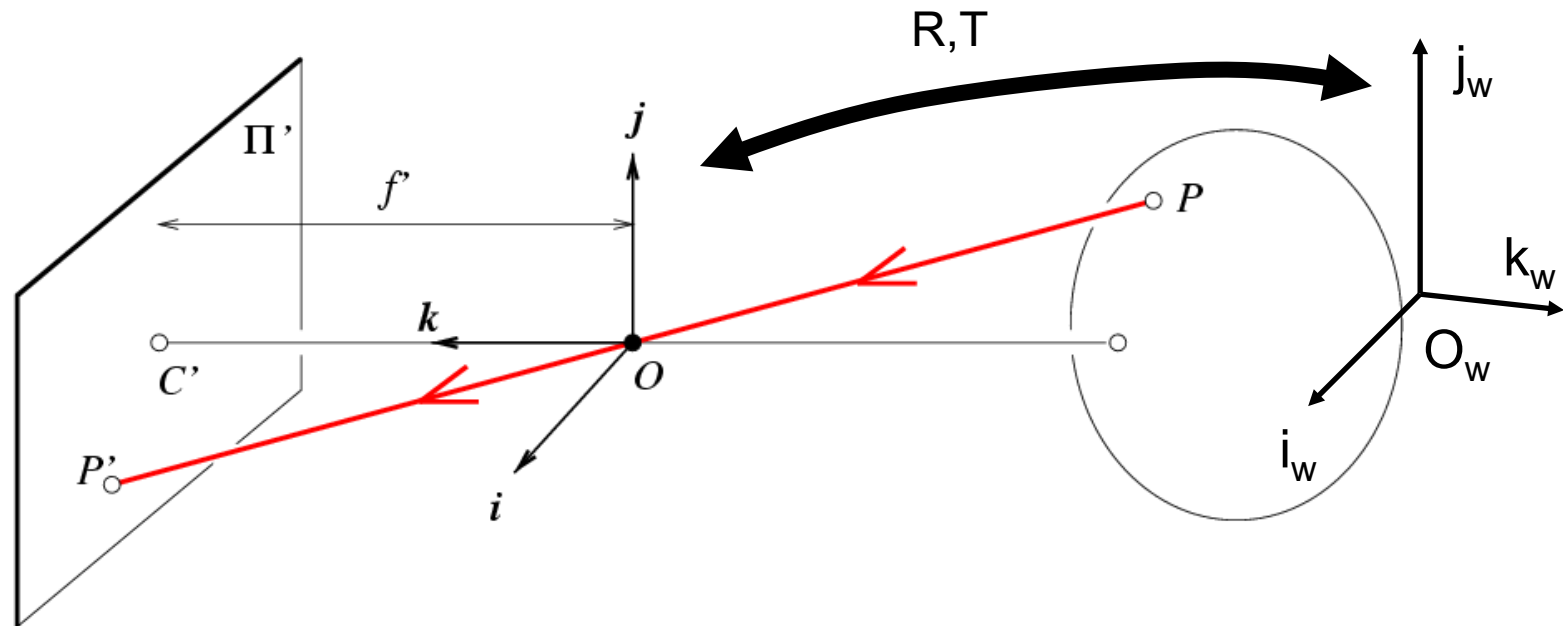
$$k \begin{bmatrix} x \\ y \\ w \end{bmatrix} = \begin{bmatrix} kx \\ ky \\ kw \end{bmatrix} \Rightarrow \begin{bmatrix} \frac{kx}{kw} \\ \frac{ky}{kw} \\ \frac{kw}{kw} \end{bmatrix} = \begin{bmatrix} \frac{x}{w} \\ \frac{y}{w} \\ 1 \end{bmatrix}$$

Homogeneous
Coordinates

Cartesian
Coordinates

A point in Cartesian is a ray in Homogeneous

Projection matrix



$$\mathbf{x} = \mathbf{K} \underbrace{\begin{bmatrix} \mathbf{R} & \mathbf{t} \end{bmatrix}}_{\text{Extrinsic Matrix}} \mathbf{X}$$

Extrinsic Matrix

$\mathbf{x}$: Image Coordinates: $(u, v, 1)$, up to scale w

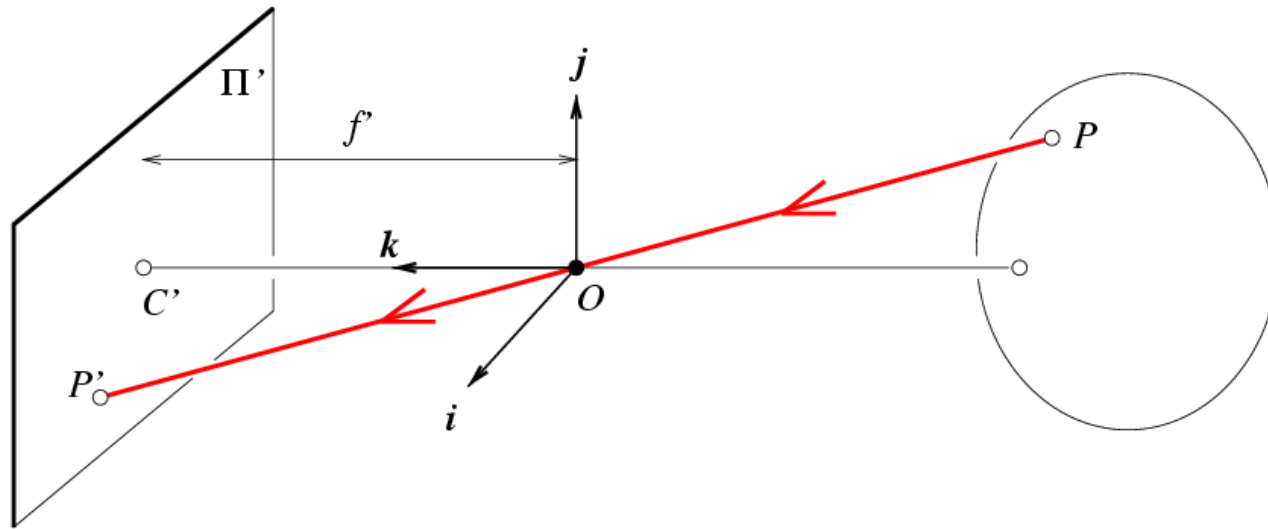
$\mathbf{K}$: Intrinsic Matrix (3x3)

$\mathbf{R}$: Rotation (3x3)

$\mathbf{t}$: Translation (3x1)

$\mathbf{X}$: World Coordinates: $(X, Y, Z, 1)$

Projection matrix



Intrinsic Assumptions

- Unit aspect ratio
- Optical center at $(0,0)$
- No skew

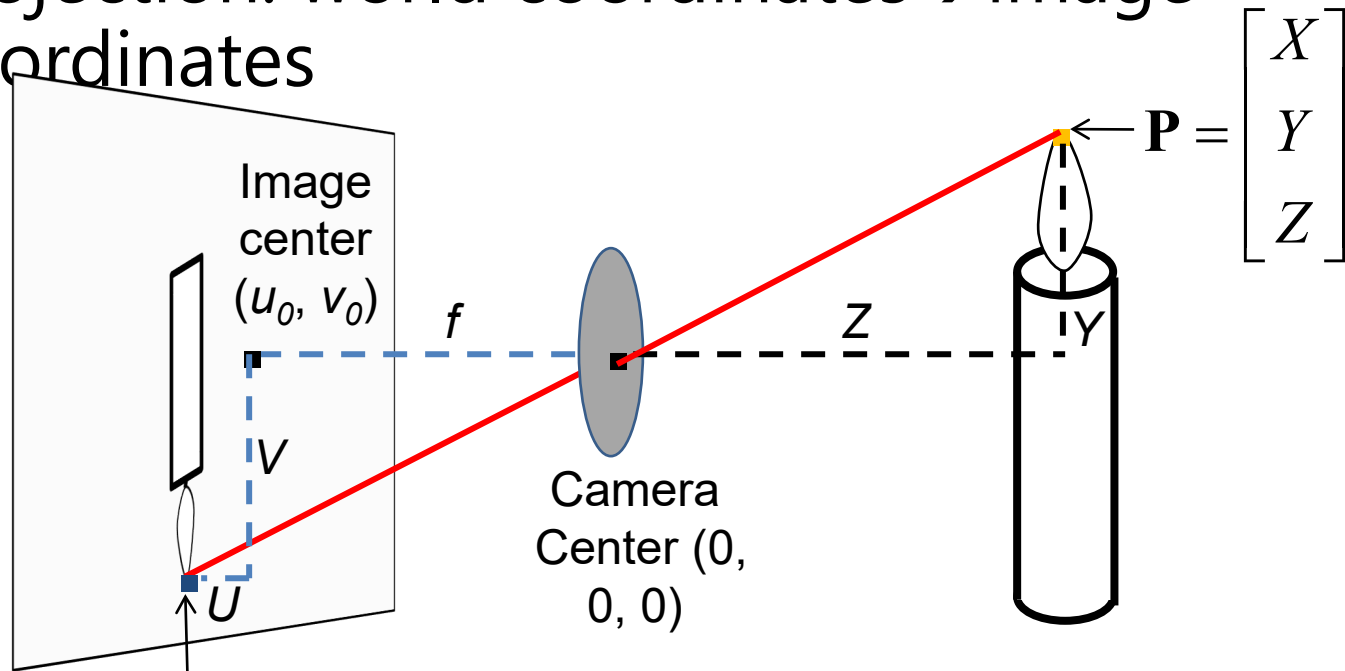
Extrinsic Assumptions

- No rotation
- Camera at $(0,0,0)$

$$\mathbf{x} = \mathbf{K} \begin{bmatrix} \mathbf{I} & \mathbf{0} \end{bmatrix} \mathbf{X} \Rightarrow {}^w \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = \begin{bmatrix} f & 0 & 0 & 0 \\ 0 & f & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

$\mathbf{K}$

Projection: world coordinates $\rightarrow$ image coordinates



$$\mathbf{p} = \begin{bmatrix} U \\ V \end{bmatrix}$$

$\mathbf{p}$ = distance from
image center

$$U = -X * \frac{f}{Z}$$

$$V = -Y * \frac{f}{Z}$$

Remove assumption known optical center

Intrinsic Assumptions

- Unit aspect ratio
- No skew

Extrinsic Assumptions

- No rotation
- Camera at (0,0,0)

$$\mathbf{x} = \mathbf{K} \begin{bmatrix} \mathbf{I} & \mathbf{0} \end{bmatrix} \mathbf{X} \Rightarrow w \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = \begin{bmatrix} f & 0 & u_0 & 0 \\ 0 & f & v_0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

Remove assumption: square pixels

Intrinsic Assumptions

- No skew

Extrinsic Assumptions

- No rotation
- Camera at (0,0,0)

$$\mathbf{x} = \mathbf{K} \begin{bmatrix} \mathbf{I} & \mathbf{0} \end{bmatrix} \mathbf{X} \rightarrow w \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = \begin{bmatrix} \alpha & 0 & u_0 & 0 \\ 0 & \beta & v_0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

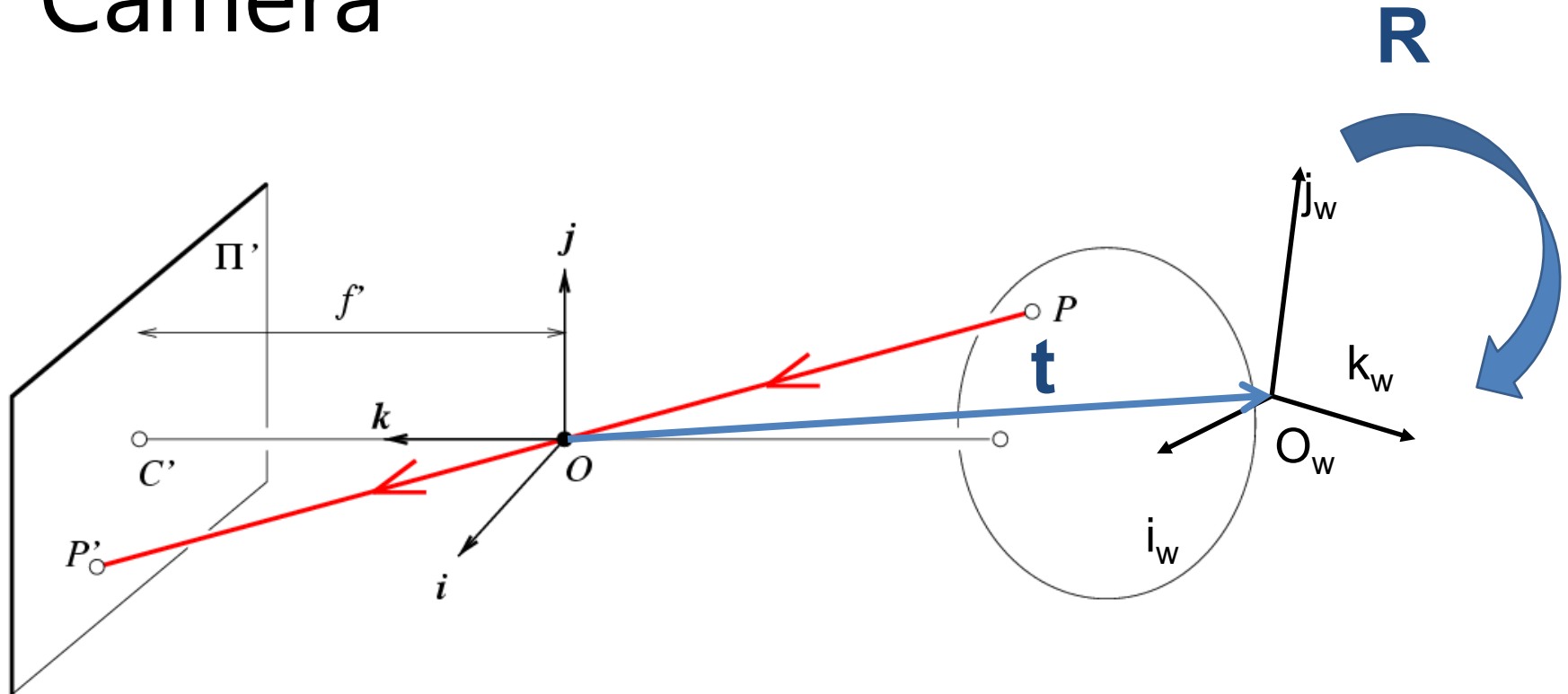
Remove assumption: non-skewed pixels

Intrinsic Assumptions Extrinsic Assumptions

- No rotation
- Camera at (0,0,0)

$$\mathbf{x} = \mathbf{K} \begin{bmatrix} \mathbf{I} & \mathbf{0} \end{bmatrix} \mathbf{X} \rightarrow w \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = \begin{bmatrix} \alpha & s & u_0 & 0 \\ 0 & \beta & v_0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

Oriented and Translated Camera



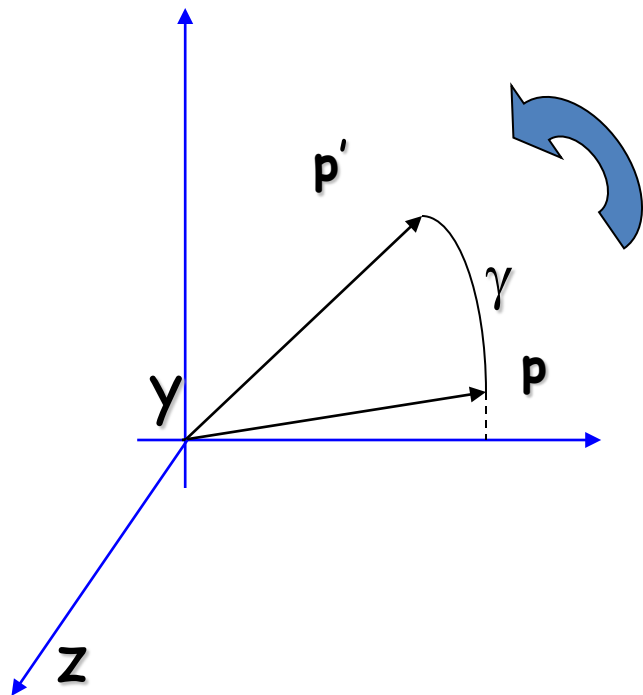
Allow camera translation

Intrinsic Assumptions Extrinsic Assumptions
• No rotation

$$\mathbf{x} = \mathbf{K} \begin{bmatrix} \mathbf{I} & \mathbf{t} \end{bmatrix} \mathbf{X} \quad \Rightarrow \quad w \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = \begin{bmatrix} \alpha & 0 & u_0 \\ 0 & \beta & v_0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & t_x \\ 0 & 1 & 0 & t_y \\ 0 & 0 & 1 & t_z \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

3D Rotation of Points

Rotation around the coordinate axes, **counter-clockwise**:



$$R_x(\alpha) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \alpha & -\sin \alpha \\ 0 & \sin \alpha & \cos \alpha \end{bmatrix}$$

$$R_y(\beta) = \begin{bmatrix} \cos \beta & 0 & \sin \beta \\ 0 & 1 & 0 \\ -\sin \beta & 0 & \cos \beta \end{bmatrix}$$

$$R_z(\gamma) = \begin{bmatrix} \cos \gamma & -\sin \gamma & 0 \\ \sin \gamma & \cos \gamma & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Allow camera rotation

$$\mathbf{x} = \mathbf{K}[\mathbf{R} \quad \mathbf{t}] \mathbf{X}$$



$$w \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = \begin{bmatrix} \alpha & s & u_0 \\ 0 & \beta & v_0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} r_{11} & r_{12} & r_{13} & t_x \\ r_{21} & r_{22} & r_{23} & t_y \\ r_{31} & r_{32} & r_{33} & t_z \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

Degrees of freedom

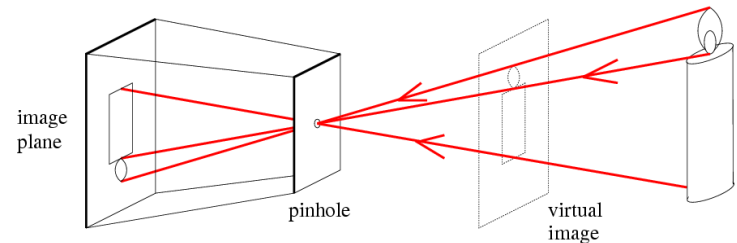
$$\mathbf{x} = \mathbf{K} \begin{bmatrix} \mathbf{R} & \mathbf{t} \end{bmatrix} \mathbf{X}$$



$$w \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = \begin{matrix} \overset{5}{\begin{bmatrix} \alpha & s & u_0 \\ 0 & \beta & v_0 \\ 0 & 0 & 1 \end{bmatrix}} \end{matrix} \begin{matrix} \overset{6}{\begin{bmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \\ r_{31} & r_{32} & r_{33} \end{bmatrix}} \end{matrix} \begin{bmatrix} t_x \\ t_y \\ t_z \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

A quick summary

- Pinhole camera model and camera projection matrix
- Homogeneous coordinates



$$\mathbf{x} = \mathbf{K} \begin{bmatrix} \mathbf{R} & \mathbf{t} \end{bmatrix} \mathbf{X}$$

$$(x, y) \Rightarrow \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

Action points

- Read the guidelines in this slides or on Canvas carefully. Feel free to ask Qi if it is not clear to you.
- Download the textbook if you haven't done so.
- Warm up with Python. Join Lab0 for help if necessary.
- Warm up with math.
- Get familiar with Piazza
- See you on Thursday

Disclaimer

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